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Statistical and Functional Representations of the Pattern of Auroral Energy Flux, Number Flux, and Conductivity

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The Hardy et al. (1985) global patterns of the integral energy flux and average energy of precipitating auroral electrons are used to determine the global pattern of the electron-produced, height-integrated Hall and Pedersen conductivities. The conductivities were determined in spatial bins in magnetic local time (MLT)-corrected geomagnetic latitude (CGL) coordinates for all MLTs and for CGLs greater than 50° and for seven levels of activity as measured by Kp. The conductivities vary smoothly with latitude and MLT typically having a single peak in latitude within the auroral oval at any MLT. On the nightside the two conductivities increase with increasing Kp. The largest conductivities are found near midnight, where the peak value of the Pedersen (Hall) conductivity varies from 3.09 (4.05) mhos to 12.5 (25.9) mhos as Kp varies from 0 to $\geq 6-$. The peak conductivity decreases with MLT away from midnight with the lowest peak values found postnoon. At noon and on much of the morning side of the oval the Pedersen and Hall conductivities increase for Kp up to 2 and then decrease for higher Kp. The highest ratios of the Hall to Pedersen conductivity are on the morning side of the oval and at noon. The peak conductivities on the dayside are significant compared to the conductivities produced by solar radiation at all seasons of the year. The global maps of the integral energy flux, integral number flux, and height-integrated Hall and Pedersen conductivities at each level of Kp were fit using both spherical harmonic and Epstein functions. The Epstein functions were found to reproduce better the original maps. At $Kp = 2$ the distribution of differences between the Epstein function fit and the original data is roughly symmetric about zero with a full width at half maximum of 16 (20)% for the Pedersen (Hall) conductivity and 32 (40)% for the integral energy (number) flux. The distribution of differences broadens with increasing and decreasing activity.

1. INTRODUCTION

In recent years there have been a number of studies modeling the ionosphere and inner magnetosphere and the processes coupling the two regions. This work has included both computer simulations [*Kamide and Matsushita*, 1979*a*, *b*; *Harel et al.*, 1981*a*, *b*; *Spiro et al.*, 1981; *Wolf et al.*, 1982; *Spiro and Wolf*, 1984] and analytic models [*Siscoe*, 1982; *Fontaine and Blanc*, 1983; *Senior and Blanc*, 1984]. In such modeling the precipitating electrons play a critical role since (1) they are a measure of the rate of loss of electrons from the inner magnetosphere into the ionosphere, (2) they operate as the primary carrier for the upward directed, field-aligned currents, and (3) they produce large changes in the ionospheric Hall and Pedersen conductivities that in turn have a major effect on the high-latitude electric field pattern [*Wolf*, 1970; *Kamide and Richmond*, 1982] and on the rate of energy dissipation due to Joule heating [*Banks et al.*, 1981; *Vickrey et al.*, 1982; *Ahn et al.*, 1983*a*, *b*; *Baumjohann and Kamide*, 1984].

Because of the importance of the electron morphology, several studies have been completed to determine the global pattern of auroral electron precipitation [*Riedler*, 1972; *McDiarmid et al.*, 1975; *Spiro et al.*, 1982; *Hardy et al.*, 1985]. In these studies, average characteristics of the precipitating electrons were determined using large quantities of data from satellite-borne particle detectors. The average characteristics were determined in elements of a spatial grid in geomagnetic coordinates with separations made according to the level of geomagnetic activity as measured by Kp or AE.

Incoherent backscatter radar data have been used, both alone and with satellite data, to determine the level of conductivity in the auroral zone and its dependence on the intensity and average energy of auroral precipitation and the location of the region 1 and 2 current sheets [*Brekke et al.*, 1974; *Banks et al.*, 1981; *Vickrey et al.*, 1981, 1982; *Vondrak and Robinson*, 1985; *Robinson et al.*, 1985, 1987]. These studies have shown that there are large latitudinal, local time, and magnetic activity variations in the conductivities with similar large variations in the relative levels of direct particle heating and Joule heating. This work has also shown that there is a functional dependence of the height-integrated Hall and Pedersen conductivities on the energy flux and characteristic energy of the precipitating electrons.

The relationship between the characteristic energy and energy flux of the precipitating electrons and the height-integrated Hall and Pedersen conductivities was used by *Spiro et al.* [1982] together with their global determination of the average pattern of electron precipitation to calculate the global pattern of conductivities. The patterns derived by Spiro et al. were later fit to a functional form to facilitate their use in models [*Reiff*, 1984; *Simons et al.*, 1985]. In addition, there have been several studies that have specified the global pattern of conductivities by explicitly calculating the ionization produced by average spectra [*Wallis and Budzinski*, 1981], by inferring the conductivity using ground-based magnetometer data [*Ahn et al.*, 1983*a*], or by using optical data [*Craven et al.*, 1984].

In this study we use the global electron precipitation pat-

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terns of *Hardy et al.* [1985] to derive the global Hall and Pedersen conductivities following the same procedures as *Spiro et al.* [1982]. These patterns, along with those of the integral number flux and integral energy flux of the electrons, are then fit to simple functional forms. The chief advantage of the present analysis over that in earlier work is that a larger data set was available to determine the precipitating electron distribution which allowed a finer binning in magnetic local time and activity and a more accurate determination of the necessary mean values. This paper is divided into five sections. In section 2 we discuss the methods for calculating the conductivities. In section 3 we present the global conductivity maps. Section 4 describes the fitting technique and the results. Section 5 is the discussion and conclusions.

2. TECHNIQUE FOR CONDUCTIVITY CALCULATIONS

The conductivities and functional representation presented in this paper are derived from the results of the statistical study of the pattern of auroral electron precipitation published previously [*Hardy et al.*, 1985]. *Hardy et al.* [1985] determined the average electron spectrum as a function of corrected geomagnetic latitude (CGL) and magnetic local time (MLT) for seven levels of activity as determined by *Kp*. At each level of activity the high-latitude region was separated into 30 zones in CGL between 50° and 90° and 48 one-half-hour zones in MLT. The zones in latitude were 2° wide between 50° and 60° and 80° and 90° and 1° wide between 60° and 80° latitude. In activity, the separation was by whole values of *Kp*, i.e., one determination for $Kp = 0, 0+$, one for $Kp = 1-, 1, 1+$, etc., up to $Kp = 5-, 5, 5+$. One final separation was made for all cases with $Kp \geq 6-$. In total, 14.1 million spectra from the SSJ/3 detectors on the DMSP F2 and F4 satellites and the CRL-251 detector on P78-1 were used in the study. These detectors measured the flux of electrons in 16 channels over the energy range from 50 eV to 20 keV (for more information on the detectors, see *Hardy et al.* [1979]). In the study, in each spatial element, and at each level of activity the average value in each of the 16 energy channels was determined. The resulting average spectra were extrapolated to 100 keV. The final spectra were integrated over energy to determine the average integral number flux and the average integral energy flux of the precipitating electrons in each spatial element. The average energy was calculated by dividing the integral energy flux by the integral number flux. For a Maxwellian distribution the average energy equals 2 k.

From such determinations of the average integral energy flux and average energy the height-integrated Hall and Pedersen conductivities can be estimated. For this we used the functional relationships of *Spiro et al.* [1982] as corrected by *Robinson et al.* [1987]. The height-integrated Pedersen conductivity Σ_P is related to the electron average energy E_0 and the electron energy flux I by

$$\Sigma_P = [40E_0/(16 + E_0{}^2)]I^{0.5} \quad (1)$$

where E_0 is expressed in keV and I is in units of ergs/cm^2 s. Equation (1) gives Σ_P in mhos. The ratio of the Hall to Pedersen conductivity is given by

$$\Sigma_H/\Sigma_P = 0.45(E_0/1\ \text{keV})^{5/8} \quad (2)$$

Equations (1) and (2) were derived from the work of *Vickrey et al.* [1981]. They assume an isotropic Maxwellian distribution for the incoming electrons.

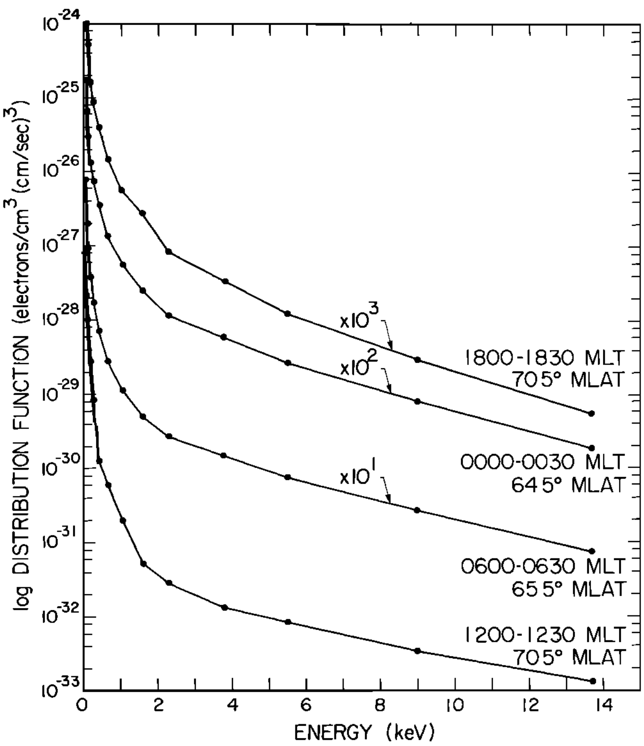

Fig. 1. The logarithm of the distribution function for electron spectra from *Hardy et al.* [1985] plotted versus energy on a linear scale. The spectra are for the $Kp = 3$ case and are taken at the approximate peak of the integral energy flux at 1800, midnight, 0600, and noon magnetic local time.

Care must be taken in using the above equations when the electron spectrum is not Maxwellian. In Figure 1 the logarithm of the distribution function is plotted versus energy for representative spectra from *Hardy et al.* [1985]. In the format of Figure 1, a Maxwellian distribution would appear as a straight line with a slope inversely proportional to the electron temperature. The distributions in Figure 1 are not Maxwellian but rather have slopes that decrease with increasing energy. As has been pointed out by *Robinson et al.* [1987], calculating the average energy for equations (1) and (2) for such distributions results in a low average energy not representative of the portion of the spectrum responsible for the *E* region ionization producing the Hall and Pedersen conductivities. They showed that using a low-energy cutoff of approximately 500 eV in the integrals to calculate E_0 avoids this problem. Such a cutoff will not underestimate the conductivity since electrons with energies less than 500 eV contribute little to the height-integrated conductivity because they produce ionization at altitudes where the collision frequency is too small to produce significant currents perpendicular to the magnetic field. Using such a cutoff, *Robinson et al.* [1987] showed excellent agreement between the conductivities calculated using equations (1) and (2) and the conductivities calculated using data from an incoherent backscatter radar. For our study, we started the integration with the channel centered at 661 eV which results in a low-energy cutoff of approximately 500 eV.

The calculated values of the Hall and Pedersen conductivities were smoothed slightly to remove small amounts of residual noise and statistical fluctuations in the data. A simple linear smoothing was used: the value of the conductivity in each spatial element was recalculated from

$$a_{i,j} = (3a_{i,j} + a_{i,j+1} + a_{i,j-1} + a_{i+1,j} + a_{i-1,j})/7 \quad (3)$$

TABLE 1. Midnight and Morning Maximum Hall and Pedersen Conductivities and Their Locations

Kp	Σ_P	Latitude, deg	Σ_H	Latitude, deg	MLT
		Nightside Conductivity Maxima			
0	3.09	68	4.05	68	2330–2400
1	4.56	68	6.41	68	2330–2400
2	7.02	66	12.7	66	0130–0200
3	8.88	66	17.6	65	0100–0130
4	8.60	66	17.0	66	0000–0030
5	...	...	...	...	...
≥6−	12.5	63	25.9	62	0100–0130
		Morningside Conductivity Maxima			
0	2.33	68	3.34	67	0300–0330
1	4.38	67	7.09	67	0300–0330
2	...	...	...	...	...
3	7.65	66	17.7	66	0400–0430
4	7.98	66	19.5	66	0430–0500
5	8.86	66	16.6	66	0430–0500
≥6−	12.7	66	24.8	65	0400–0430

where $a_{i,\,j}$ refers to the conductivity measured in the ith latitude bin and the jth local time bin. Three iterations of the smoothing were applied to the data.

3. CONDUCTIVITY RESULTS

In Plates 1 and 2 the calculated height-integrated Hall and Pedersen conductivities are displayed as polar color spectrograms for each of the seven levels of activity in a magnetic local time-corrected geomagnetic latitude coordinate system. The main features of the variations are as follows:

1. The region of elevated conductivity is confined to the annular region of auroral precipitation centered several degrees away from the geomagnetic pole toward the nightside and towards dawn. With increasing activity the annular region moves equatorward and the level of conductivity within the annulus systematically increases. The conductivity, in general, has a single maximum at each magnetic local time and varies smoothly with both latitude and local time.

2. The higher levels of both Hall and Pedersen conductivities occur in horseshoe or C-shaped regions. Except for the $Kp = 0$ case the patterns of both Hall and Pedersen conductivity tend to be roughly symmetric about a postnoon postmidnight meridian. The one exception is the $Kp = 5$ case where the highest conductivity levels are found toward the dawn and dusk meridians and there is a decrease in the maximum conductivity toward midnight. That this is a real effect is unclear. Only approximately 5% of the data used occurred during periods with $Kp = 5$. It is possible that the effect is a statistical artifact of the relatively low number of samples used to construct the averages for this case. We are currently repeating the electron study using data from the newer SSJ/4 spectrometers on the DMSP F6 and F7 to check if this feature persists.

3. For both the Pedersen and Hall conductivity there are consistently two local maxima at each activity level on the nightside of the oval: one near midnight (from 2330 to 0130 MLT) and the other at local times in the early morning (from 0300 to 0500 MLT). A premidnight maximum is seen for $Kp = 4$ and 5. The values of the midnight and morning maximum Hall and Pedersen conductivities and their locations are listed in Table 1. The maximum near midnight is seen for all cases except $Kp = 5$ and the early morning maximum is seen at all activities except $Kp = 2$. In general, the maximum near midnight at each Kp is slightly larger than the maximum in the early morning. For both sets of maxima the value of the conductivity increases with increasing activity. For the midnight maxima, the increase in the Pedersen (Hall) conductivity from $Kp = 0$ to $Kp \geq 6-$ is a factor of 4.04 (6.40). For the early morning maxima the increase is a factor of 5.45 (7.42). The ratio of the Hall to Pedersen conductivities at the maximum increases with increasing activity up to $Kp = 3$ but is roughly constant at higher values. This reflects the behavior of the average electron spectra in the nightside region, which harden with increasing Kp up to $Kp = 3$ but maintain approximately constant average energies at higher activities [*Hardy et al.*, 1985]. The observation of a double maximum on the evening side of the oval has been previously reported by *Wallis and Budzinski* [1981].

TABLE 2. Noon Conductivity Maxima

Kp	Σ_P	Latitude, deg	Σ_H	Latitude, deg
0	2.15	73	3.80	73
1	2.35	73	4.72	72
2	2.45	73	5.94	72
3	2.26	73	5.81	72
4	2.18	72	4.83	71
5	1.70	71	3.75	70
≥ 6−	1.94	72	3.04	66

4. Within the annular region both the Hall and Pedersen conductivities systematically decrease with magnetic local time from the nightside toward noon. The locations and values of the peak Hall and Pedersen conductivities along the noon meridian are listed in Table 2. Both conductivities are seen to increase slightly from $Kp = 0$ to $Kp = 2$ and to decrease for higher Kp. The location of the maxima is confined between 70° and 73° with a single exception. The ratio of the peak Hall (Pedersen) conductivity at midnight to that at noon increases with increasing activity from 1.3 (1.4) to 6.8 (7.7) from $Kp = 0$ to $Kp \geq 6-$. The decrease in the dayside conductivity for Kp greater than 2 is the result of the decrease in the flux of high-energy electrons that precipitate at noon for these activities, as discussed by *Hardy et al.* [1985].

5. The lowest peak conductivities are seen postnoon. In Table 3 we have listed the values and locations of the minimum peak value in the conductivities. As for the noon values, the minimum values are reasonable constant: increasing slightly up to $Kp = 3$ and flat or decreasing at higher activities. For the Hall conductivity the minima tend to be at lower latitudes and later local times than for the Pedersen conductivity minima. Since the values at the minimum show little variation, the ratio of the maximum to minimum peak value within the oval increases with increasing activity. For the Ped-

TABLE 3. Minimum Peak Conductivity

Kp	Σ_P	Latitude, deg	MLT	Σ_H	Latitude, deg	MLT
0	1.04	72	1600–1630	1.01	70	1900–1930
1	1.23	74	1430–1500	1.82	71	1630–1700
2	1.44	73	1330–1400	2.70	72	1700–1730
3	1.46	73	1300–1330	2.19	74	1530–1600
4	1.19	75	1400–1430	1.80	72	1530–1600
5	1.35	71	1230–1300	2.21	71	1430–1500
≥ 6−	0.936	73	1300–1330	1.47	72	1430–1500

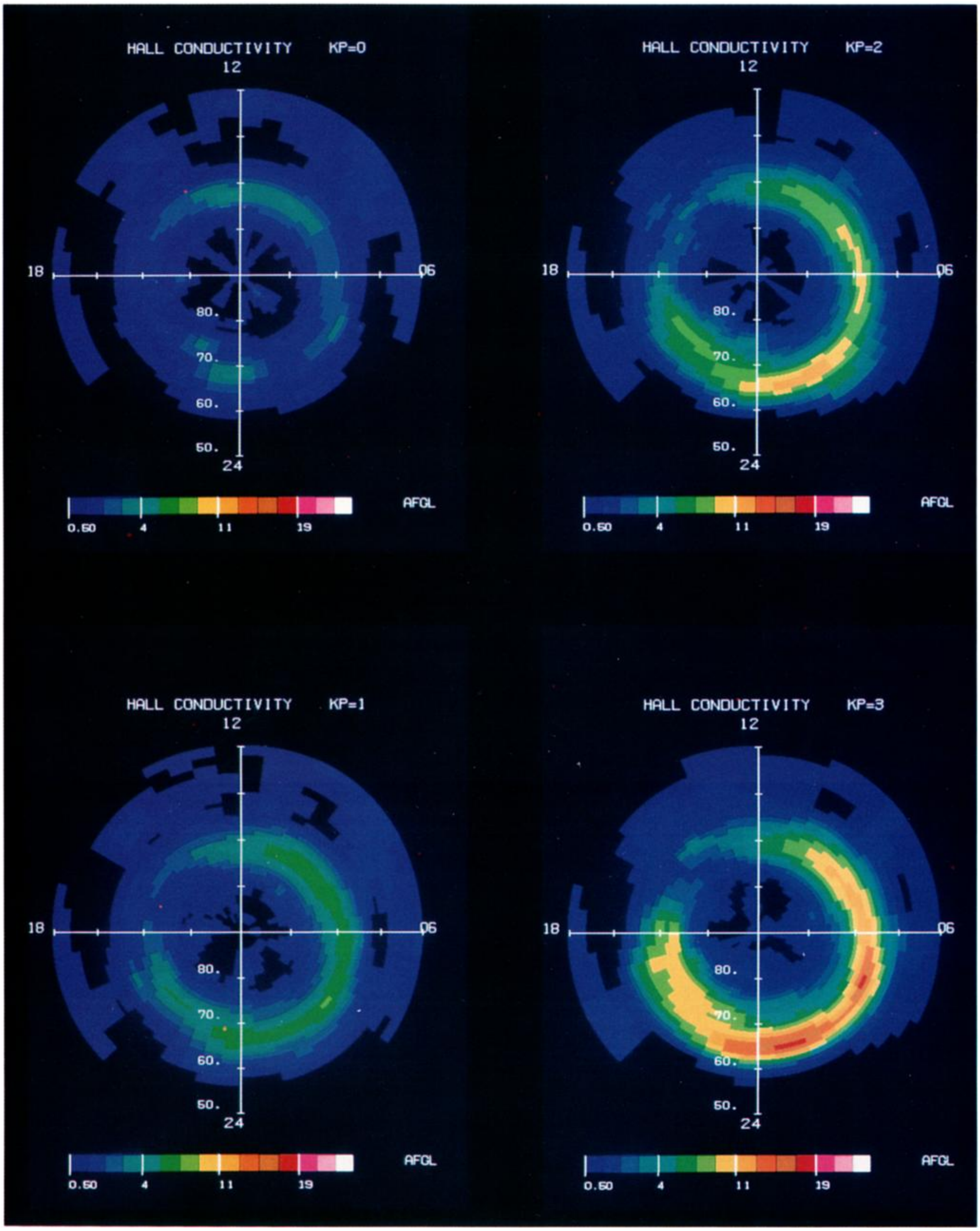

Plate 1. Polar color spectrograms of the Hall conductivity for each of seven level of Kp. The values of the conductivity are plotted in a magnetic local time-corrected goemagnetic latitude coordinates for latitudes above 50°. The 15 color levels correspond to conductivities of 0.5, 1.0, 2.0, 3.0, 4.0, 5.0, 7.0, 9.0, 11.0, 13.0, 15.0, 17.0, 19.0, 21.0, >23.0mhos.

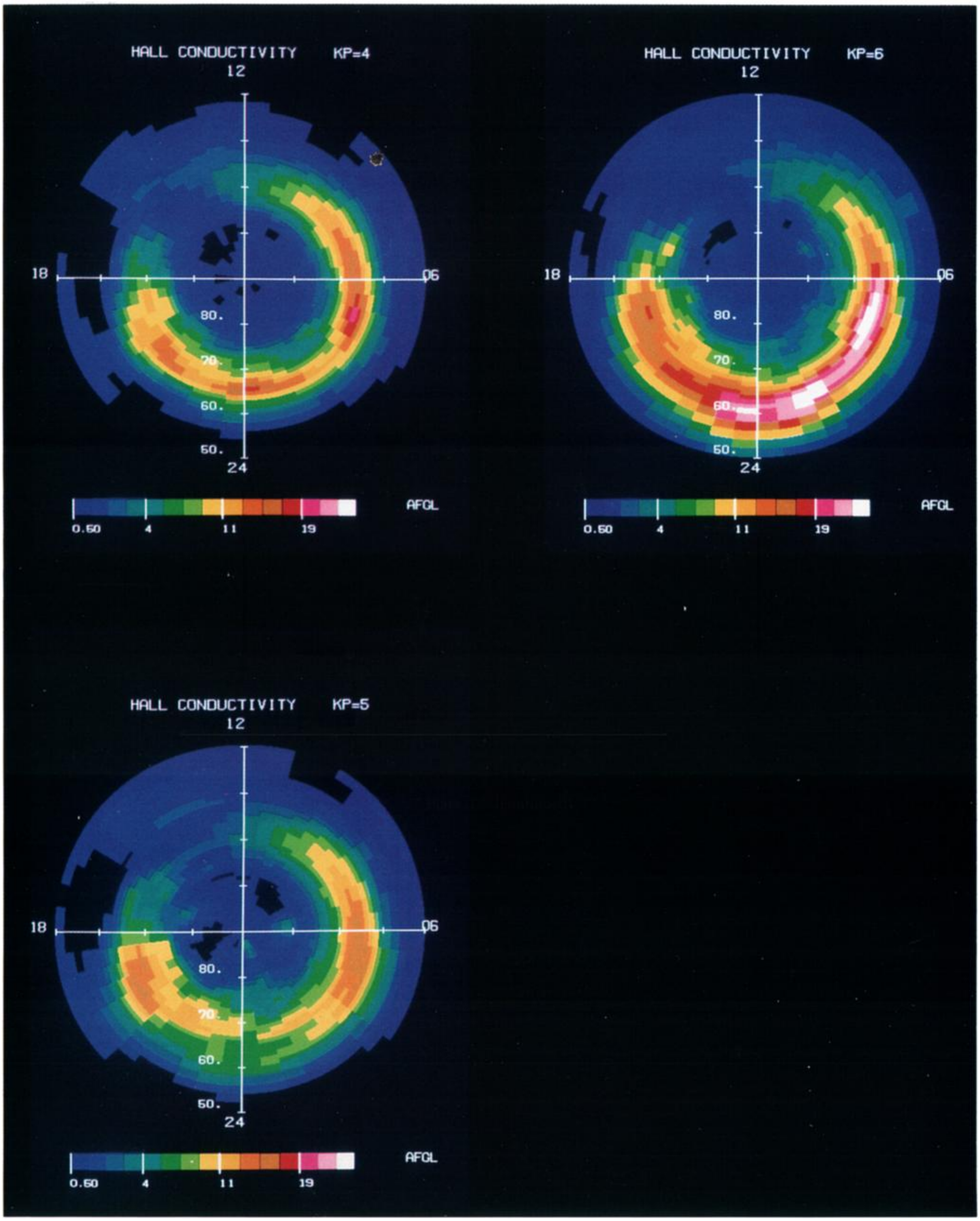

Plate 1. (continued)

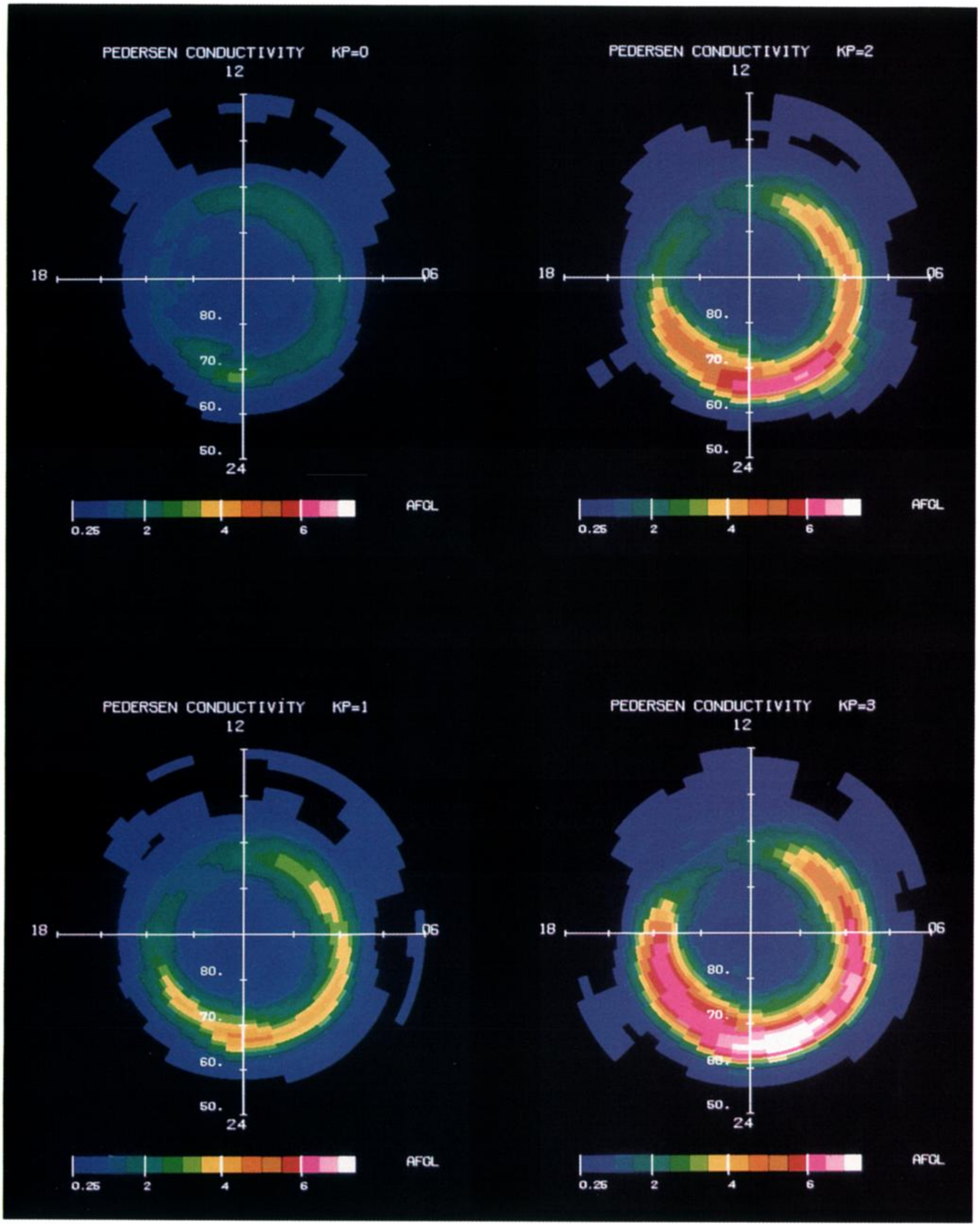

Plate 2. Same as Plate 1 for the Pedersen conductivities. The 16 color levels correspond to conductivities of 0.25, 0.50, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5, 6.0, 7.0, 8.0, >9.0 mhos.

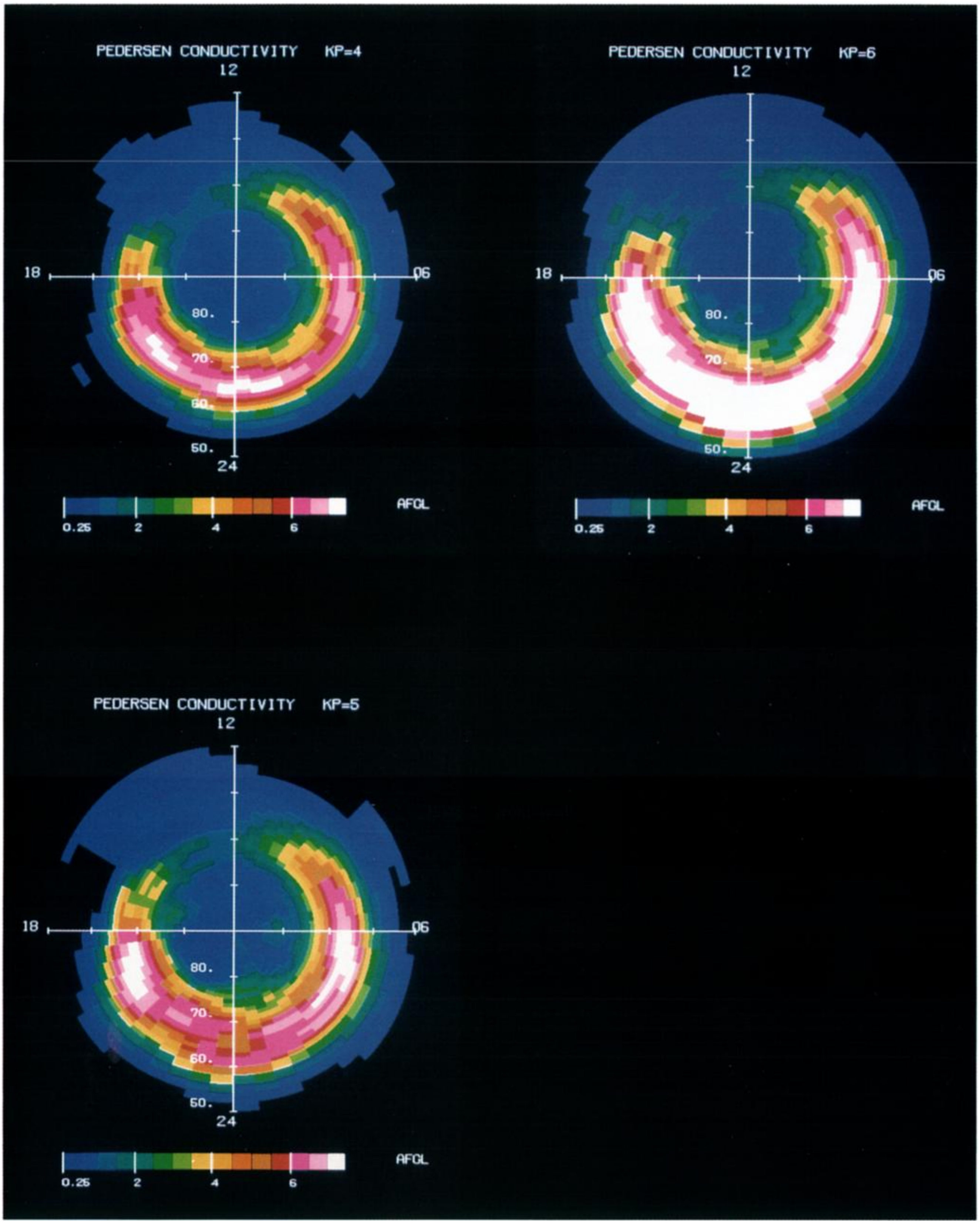

Plate 2. (continued)

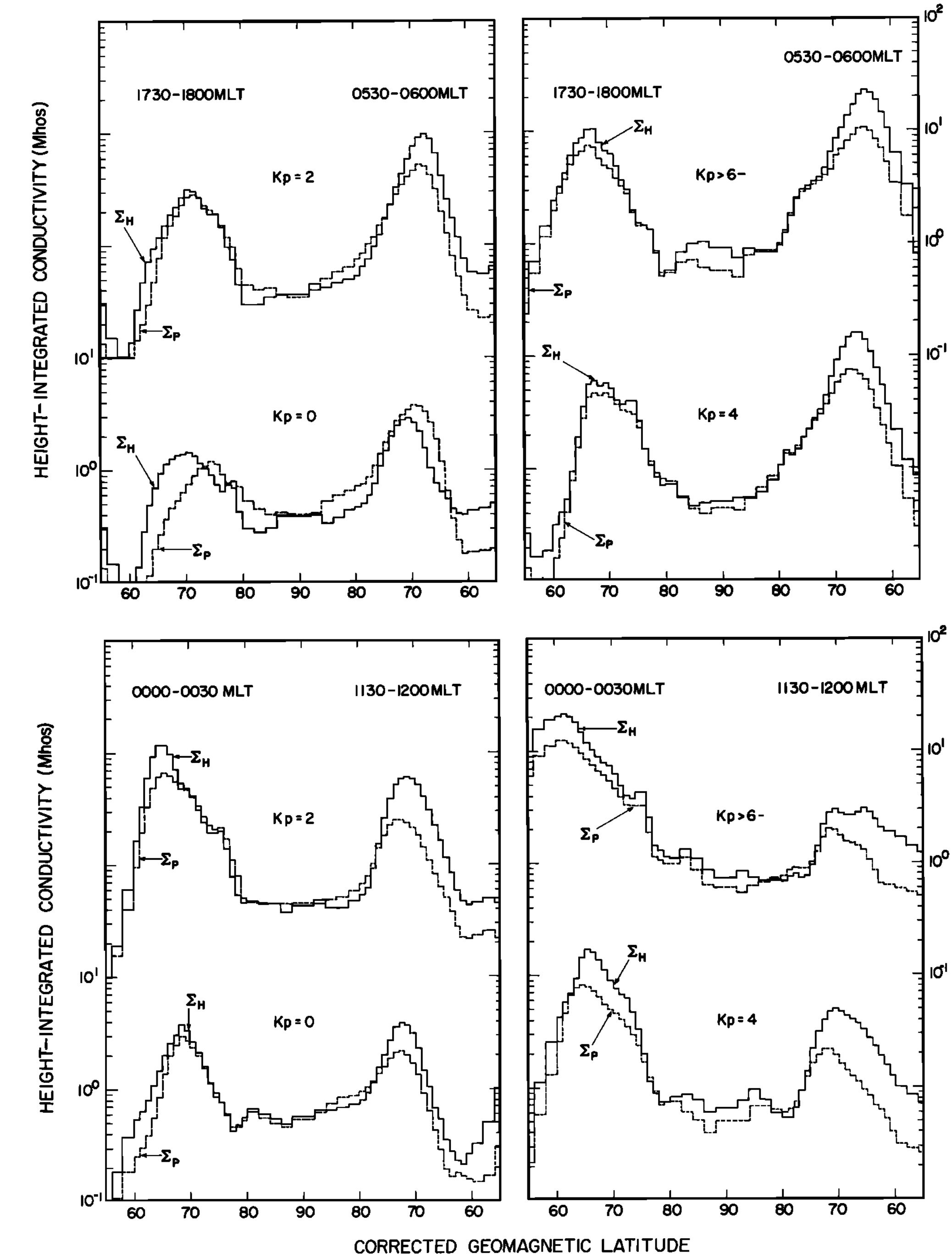

Fig. 2. Profiles of the Hall and Pedersen conductivities along the noon-midnight and dawn-dusk meridians for $Kp = 0$, 2, 4, and $\geq 6-$.

ersen conductivity the ratio varies from 2.97 to 13.6 and for the Hall conductivity from 4.00 to 18.6 from $Kp = 0$ to $Kp \geq 6-$.

A comparison of the latitudinal profiles for the height integrated Hall and Pedersen conductivities are shown in Figure 2. Here the profiles have been plotted for the dawn-dusk and noon-midnight meridians for the $Kp = 0$, 2, 4, and $\geq 6-$ cases. Figure 2 illustrates the following:

1. Along a particular meridian the conductivity profile exhibits two local maxima with an extended minimum in the polar cap and low values at low latitudes. The shape of the conductivity profile remains roughly constant at all levels of activity above $Kp = 0$ at a specific local time. Only the position of the maximum and the overall intensity vary. For both conductivities, the profiles typically are not symmetric in latitude. Except near noon, they have a steeper slope on the equatorward side of the maximum value than on the poleward side. In general, except at noon, the maximum moves to lower latitudes, and the intensity increases with increasing activity. At noon the maximum value of the Hall and Pedersen conductivities increases for low activities but then decreases for higher activities in agreement with Table 2. The conductivity in the polar cap is relatively uniform with values in the range from 0.3 to 1.0 mhos and shows no significant variation with

activity. At 85° the average value of the Pedersen (Hall) conductivity is approximately 0.57 (0.55) mhos at all levels of activity.

2. At 1800 MLT the Hall and Pedersen conductivities are approximately equal at all latitudes and all activities above $Kp = 0$. At the peak of the conductivities in latitude the ratio does exceed 1.0, but even for the $Kp \geq 6-$ case the ratio does not exceed 1.5. The approximate equality of the Hall and Pedersen conductivities at 1800 reflects the relatively low average energy of the electrons on the eveningside of the oval and is a general feature of the evening side oval for $Kp > 0$. The low average energies on the eveningside arise because the trajectory for low-energy electrons in the inner magnetosphere under the combined action of convection and gradient and curvature drifts primarily threads the evening flank of the magnetosphere before intersecting the magnetopause [*Ejiri*, 1978].

3. The largest ratios of Hall to Pedersen conductivity lie at or near the conductivity peak in the midnight and morning sectors (ratios up to 2) and below the peak on the dayside (ratios exceeding 3). These higher ratios reflect the harder electron spectra in the morningside diffuse aurora and the hardening of these spectra toward noon [*Hardy et al.*, 1985]. Between midnight and 0600 the ratio decreases with latitude above and below the peak, and there is a significant range of latitudes poleward of the maximum for which the ratio of Hall to Pedersen conductivities is approximately 1 or less. This decrease in the ratio for latitudes above the peak we attribute primarily to the conductivity resulting from the colder electrons of the boundary plasma sheet and cleft or cusp toward the poleward edge of the oval [*Winningham et al.*, 1975]. It should be noted that in this region, the average spectra are a composite of the cold spectra of the boundary regions and the accelerated spectra in the discrete arcs and the conductivity is calculated for such a composite.

4. The $Kp = 0$ conductivity patterns are significantly different from those at all higher levels of activity. On the eveningside of the oval the Hall conductivity peaks 4° equatorward of the Pedersen conductivity and the ratio of Hall to Pedersen conductivities has values exceeding 3.0. At midnight the peaks coincide and no such large ratios are observed. At 0600 the peak Pedersen conductivity exceeds the peak Hall conductivity, and the Hall to Pedersen ratios are as small as 0.42. The smallest ratios are found equatorward of the peak values for both conductivities. These effects result from a reversal of the pecipitation pattern seen at higher activities where the hotter electrons are observed on the morningside and the cooler electrons on the eveningside. For $Kp = 0$, spectra with average energies exceeding 10 keV are observed at 1800, and the spectra at 0600 have average energies of less than 1 keV.

The dependence of the Hall and Pedersen conductivity ratio on the average energy is explicitly shown in Figure 3. Here we have again plotted the Hall and Pedersen conductivity profiles along the noon-midnight meridian together with the profiles of the average energy and integral energy flux (both calculated for energy channels starting at 661 eV). The ratio of the Hall to Pedersen conductivity is seen to follow the average energy. The smallest ratios are at midnight, poleward and equatorward of the peak integral energy flux, at noon well poleward of the peak integral energy flux, and in the polar cap. In all these regions the value of the average energy is 2 keV or less.

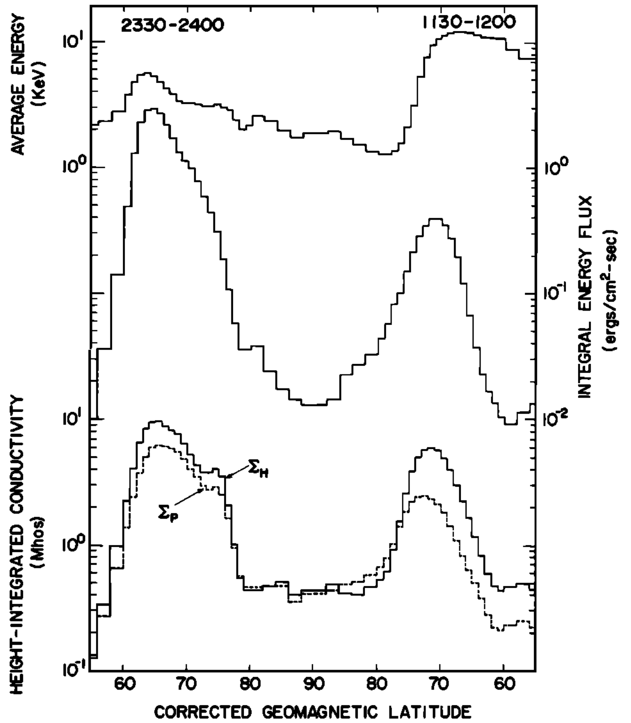

Fig. 3. The Hall and Pedersen conductivities along the noon-midnight meridian plotted along with the energy flux and average energy for precipitating electrons with energies greater than 661 eV for $Kp = 3$.

The larger ratios at noon reflect the greater than 10-keV average energies found there.

The level of the Pedersen conductivity at noon, and generally to the poleward edge of the oval, may be underestimated since the calculation of the Pedersen conductivity does not take into account the energy flux due to low-energy electrons precipitating in the cusp or cleft. Near the Pedersen conductivity maximum at noon such low-energy electrons can account for 50% of the total energy flux. The contribution of this energy flux to the conductivities is most likely small since such low-energy electrons will deposit their energy and produce ionization in the F region where transport plays an important role in determining the equilibrium plasma density and where the collision frequency is low.

4. Functional Representation of the Patterns of Auroral Flux and Conductivity

As shown above, the variation in the integral number flux, integral energy flux, and conductivities is smooth and regular in both magnetic local time and corrected geomagnetic latitude. Further, for most levels of activity the shape of the variation of the quantities in latitude, at a given local time, varies only slightly with activity, while the overall intensity increases or remains constant. These regularities in the patterns make it possible to fit the patterns to a simple functional form. The advantage of such a functional fit is that the pattern can be reproduced without storing the full set of average values such that the patterns can be more easily included in global models.

The patterns of integral energy flux, integral number flux, and Hall and Pedersen conductivities were fit for all seven levels of activity. The values of the integral energy flux and integral number flux that were fit were calculated over the entire electron spectrum. For the conductivities, only the energy channels starting at 661 eV were used. The average energy was not fit since it can be calculated from the integral energy flux and integral number flux. The patterns were represented using both a seventh-order spherical harmonic fit and using fits to Epstein transition functions. Gaussian distributions, as were used by *Reiff* [1984] and *Simons et al.* [1985], were not used since, as we have shown, the variation in the quantities with latitude is not symmetric about the maximum. Since the Epstein transition function was found to repeat better the original data, only the Epstein function fitting procedure will be described and the results listed.

The Epstein transition function used to fit the data is of the form,

$$e(h) = r + S_1(h - h_0) + (S_2 - S_1) \cdot \ln \{[1 - S_1/(S_2 e^{-(h-h_0)})]/[1 - (S_1/S_2)]\} \qquad (4)$$

where

h corrected geomagnetic latitude, in degrees;
r maximum value of the quantity e being fit;
h_0 latitude for the maximum value of e at one MLT;
S_1 slope of the curve for $h \ll h_0$;
S_2 slope of the curve for $h \gg h_0$.

The function has the characteristic that for latitudes much less than h_0 the slope of the curve approaches S_1, and for latitudes much greater than h_0 the slope of the curve approaches S_2. This results in a better representation of the asymmetric profiles than the Gaussian fits used by *Reiff* [1984] and *Simons et al.* [1985].

The fitting was performed as follows: For each magnetic local time and for each of the four quantities (integral number flux, integral energy flux, Hall conductivity, and Pedersen conductivity), the maximum value and the latitude of the maximum were determined, and a least squares fit was performed for points well below and above the peak separately to find the optimum value for S_1 and S_2, respectively. For the integral energy flux and integral number flux the fits were performed on the logarithm of the values because of the large range in the values of these quantities with latitude. For the two conductivities the fits were made to linear values. Only points above a specified minimum were used in the least squares fits. This was to exclude the values below the auroral zone and in the polar cap where the values of the four quantities are roughly constant or zero. For the integral energy flux and integral number flux the minimum was 10^6 keV or electrons/cm^2 s sr to the equatorward side of the maximum and 10^7 to the poleward side. For the conductivities, the minima were 1 mho both poleward and equatorward of the maximum.

All fits were visually inspected and in 80% of cases were found to give good results. In the remaining 20% of the cases, new ranges and peaks were manually selected to compensate for residual noise in the data, and the fitting procedure was repeated. In general, the fitting procedure gave the value of the maximum and the latitude of the maximum exactly. The fitting procedure yielded a total of 192 coefficients (four for each of 48 magnetic local time sectors) for each of the seven levels of activity and for each of the four quantities. Inspection of the coefficients showed them to be slowly varying with magnetic local time. The number of coefficients needed to represent the pattern could be reduced by expanding the MLT dependence in a Fourier series of the form,

$$a(T) = \Sigma\{C_n^a \cos [n(T/12)] + S_n^a \sin [n(T/12)]\} \qquad (5)$$

where C_n^a and S_n^a are the Fourier coefficients with the superscript a referring to each of the four values fit and T is the MLT in hours, and the functions to be represented are $a(T) = r$, h_0, S_1, or S_2. The series was taken to order 6. Thus each of the four constants of the Epstein transition function (r, h_0, S_1, and S_2) is represented by a Fourier series with 13 coefficients, and each map is represented using a total of 52 coefficients. This is to be compared to the 1440 averages determined for each map. The coefficients determined in this way for each of the four quantities are listed in Tables 4–7. Tables 4–7 contain the coefficients for the Fourier expansions that fit the MLT variation in the four constants in the Epstein function for the integral number flux, integral energy flux, Hall conductivity, and Pedersen conductivity for each of the seven Kp levels. The listings give the coefficients for the cosine and sine terms for the maximum value, the maximum latitude, and the upslope and the downslope for each of the four constants at each Kp level. These correspond to the quantities r, h_0, S_1, and S_2 in equation (4). The values can be calculated by inserting the coefficients in equation (5) to derive the constants and then inserting these constants into equation (4). The sets of 192 coefficients and the coefficients for the spherical harmonics fits are avaiable from the Air Force Geophysics Laboratory.

In using the fits, the integral energy flux and integral number flux should be set equal to 10^6 keV/cm^2 s sr and 10^6 el/cm^2 s sr, respectively, at the latitudes equatorward of the maximum where the Epstein function gives values less than these levels. Poleward of the maximum the values should be set to 10^7 for both quantities. For the conductivities the values for both Hall and Pedersen conductivities should be set equal to zero equatorward of the maximum anywhere the functions give a value less than 0; and approximately 0.55 mhos poleward of the maximum where the function gives a value less than this level.

The variation in the constants of the Epstein function with MLT and the quality of the Fourier expansion to the values is illustrated in Figures 4 and 5. In Figures 4 and 5 the variation of the four Epstein transition functions' constants for the Pedersen and Hall conductivities are plotted for $Kp = 3$ as a function of MLT. The $Kp = 3$ case is representative of the fits at the other levels of activity. In general, the variation in MLT is seen to be smooth, and the Fourier expansion provides an excellent approximation to the points. For both the Hall and Pedersen conductivities the peak value is seen slightly postmidnight. The peak value decreases with local time away from midnight with a minimum for the Pedersen (Hall) conductivity at 1300 (1500) hours MLT. The latitude of the peak value for both conductivities is lowest at midnight and highest at 1400 (1600) hours MLT for the Pedersen (Hall) conductivity. The steepest slopes equatorward of the peak ("upslope") are seen near midnight for both conductivities. These slopes decrease greatly with MLT toward noon. The slopes poleward of the peaks ("downslope") show a smaller variation with MLT and in magnitude can exceed the equatorward slopes near noon. It should be noted that one significant failing of the fit is that the second local maximum in the predawn region is not well reproduced.

TABLE 4. Epstein-Fourier Coefficients for Energy Flux

		max value	max latitude	up-slope	down-slope
K, 0	Constant	7.7896976	73.0708333	.2191803	-.0859725
	Cos 1	.0002967	-3.3538923	.0563385	-.0398150
	Cos 2	.0768187	-1.0653413	.0123579	-.0220984
	Cos 3	.0935700	1.2190125	-.0517093	-.0131578
	Cos 4	.0028059	-.5114249	-.0055988	.0168468
	Cos 5	.0147765	-.2189668	.0006074	.0119485
	Cos 6	.0021958	-.5446280	.0332549	.0146299
	Sin 1	.0462879	-1.8623847	.0125471	-.0011417
	Sin 2	.0240691	-.0758685	-.0234464	-.0234705
	Sin 3	-.0399482	.0309485	-.0165087	-.0098271
	Sin 4	-.0516997	.1384184	-.0043154	-.0043996
	Sin 5	-.0441358	.3436580	.0084612	-.0008923
	Sin 6	-.0008464	-.4349738	.0357420	.0028314
K, 1	Constant	8.0364974	71.1875000	.2490905	-.0888827
	Cos 1	.1961799	-3.4878153	.0606912	-.0293764
	Cos 2	-.0023255	-.3527452	-.0225351	-.0113256
	Cos 3	.0429868	.6444892	-.0143288	-.0089596
	Cos 4	-.0433994	-.6410611	.0232768	.0217679
	Cos 5	.0277324	.1732289	-.0054526	.0050594
	Cos 6	.0029320	-.3896202	.0075123	.0104945
	Sin 1	.0797814	-2.5027481	.0401181	.0185842
	Sin 2	-.0235719	.3886167	.0029146	-.0027588
	Sin 3	-.0169762	-.5568983	.0015539	.0201468
	Sin 4	-.0148749	.0725993	-.0051051	.0060130
	Sin 5	.0137977	-.1717545	.0189600	-.0000570
	Sin 6	.0014145	-.0497699	.0240313	-.0025187
K, 2	Constant	8.2850158	70.4062500	.2460024	-.1171000
	Cos 1	.2665715	-3.8822697	.0563624	-.0200770
	Cos 2	.0095929	-.3345333	-.0173165	.0023607
	Cos 3	.0562738	.6493035	-.0081089	-.0113432
	Cos 4	-.0444980	-.6410611	.0582433	.0152396
	Cos 5	.0108284	.2531377	.0166424	-.0013597
	Cos 6	-.0015142	-.0962376	-.0085391	.0016201
	Sin 1	.1217648	-2.4335720	.0302470	.0199753
	Sin 2	-.0444369	.0509922	-.0246440	-.0037281
	Sin 3	.0387839	-.8521281	-.0103219	-.0042125
	Sin 4	-.0061023	-.0697097	.0129462	.0030722
	Sin 5	.0427911	-.3485596	.0384384	-.0066734
	Sin 6	-.0082458	.3252729	.0147419	-.0018083
K, 3	Constant	8.4772806	69.2812500	.2753726	-.1115326
	Cos 1	.3651667	-4.0683244	.1151785	-.0220689
	Cos 2	-.1059687	-.1225232	.0234371	.0178650
	Cos 3	.0324384	-.0178301	.0314753	-.0018842
	Cos 4	-.0132120	-.6356690	.0288739	.0074899
	Cos 5	.0591799	-.0265770	-.0154872	-.0040025
	Cos 6	-.0081149	-.1759633	-.0112962	-.0021947
	Sin 1	.1096211	-2.4959098	.0372748	.0126324
	Sin 2	-.0351757	.2854165	.0397184	-.0011085
	Sin 3	.0659091	-.9594749	.0534929	-.0084360
	Sin 4	.0254995	.5512278	.0172888	-.0020236
	Sin 5	-.0133701	.0565716	-.0011013	-.0049755
	Sin 6	.0145300	.3412180	.0037713	-.0052334
K, 4	Constant	8.5195810	68.6770833	.2266293	-.1344448
	Cos 1	.4168092	-3.7241896	.0254753	-.0081620
	Cos 2	-.1601158	.7259087	-.0633029	.0008903
	Cos 3	.0078810	-.0121933	.0125143	-.0051142
	Cos 4	.0092577	-.4427162	.0234768	.0159459
	Cos 5	.0509579	.2728201	-.0247079	-.0182547
	Cos 6	-.0051431	-.0511379	-.0249497	.0008094
	Sin 1	.1571118	-2.0812021	.0033675	.0227951
	Sin 2	-.1385018	.6227584	-.0079194	-.0133774
	Sin 3	.0342856	-.7456933	.0167988	.0098810
	Sin 4	.0149269	.8020486	-.0306430	-.0192107
	Sin 5	.0042574	.1579351	-.0123330	-.0095057
	Sin 6	.0350032	-.2933826	.0294261	.0093728
K, 5	Constant	8.5256150	67.5500000	.1829103	-.1114507
	Cos 1	.3833655	-3.3575591	.0248435	.0032621
	Cos 2	-.2472626	.4815741	-.0355286	.0075126
	Cos 3	-.0221512	-1.0500534	.0331517	.0272683
	Cos 4	-.0456430	-.6766072	.0152159	.0233378
	Cos 5	.0591230	-.0542310	-.0044664	-.0131719
	Cos 6	.0175221	-.5671779	-.0065978	.0145349
	Sin 1	.0912572	-1.9088494	.0221958	.0138889
	Sin 2	-.0668920	-.3041218	.0090232	.0057275
	Sin 3	.0040225	-1.1673833	.0383684	.0295801
	Sin 4	.0106788	.4998871	.0078454	-.0184993
	Sin 5	.0228927	-.5881622	.0268375	.0124451
	Sin 6	.0010010	-.2342528	.0080741	-.0054305
K, 6	Constant	8.6809870	66.2916667	.1601541	-.1060242
	Cos 1	.5954141	-4.9456659	.0484496	-.0009483
	Cos 2	-.1751444	.5050106	-.0263287	.0041099
	Cos 3	-.0089652	-.8573961	-.0067494	.0077486
	Cos 4	.0796042	-.5881981	.0177230	.0040821
	Cos 5	.0257641	.2395331	-.0112026	-.0099643
	Cos 6	-.0060951	-.2515854	-.0211823	.0171560
	Sin 1	.1824138	-1.1950328	.0270234	-.0020533
	Sin 2	-.0593238	.9749190	-.0055590	-.0110137
	Sin 3	.0423806	-.8919033	.0197485	.0078150
	Sin 4	.0322709	.0765466	.0147867	-.0092857
	Sin 5	-.0467562	-.4742351	.0107573	.0247198
	Sin 6	.0581636	-.3604655	.0135490	-.0182895

TABLE 5. Epstein-Fourier Coefficients for Number Flux

		max value	max latitude	up-slope	down-slope
K, 0	Constant	8.1810131	75.7083333	.2091148	-.0956275
	Cos 1	-.2676399	-4.8242451	.0277581	.0438939
	Cos 2	-.0351920	-1.4782888	.0438852	-.0001422
	Cos 3	.0612769	-.0986596	.0217138	-.0138387
	Cos 4	.0101369	.4653417	-.0073231	-.0077334
	Cos 5	.0013806	.0759315	-.0096685	-.0018326
	Cos 6	-.0012810	-.0769901	-.0003772	.0011037
	Sin 1	.0589730	-.0515624	-.0076504	-.0032528
	Sin 2	-.0037701	-.3387303	.0134144	.0014625
	Sin 3	-.0146568	-.1103123	.0095689	-.0012418
	Sin 4	-.0075195	-.1943977	.0202667	.0004149
	Sin 5	-.0352166	.0058707	.0063682	.0037899
	Sin 6	-.0143828	-.0703853	.0146708	.0030647
K, 1	Constant	8.2364033	74.6041667	.2031531	-.0978087
	Cos 1	-.2463092	-5.0291482	.0023882	.0439549
	Cos 2	.0309183	-1.2117782	.0542579	-.0116659
	Cos 3	.0624215	.0734319	.0049831	-.0134314
	Cos 4	.0067655	.3755086	-.0060284	-.0031075
	Cos 5	-.0020184	-.0527220	.0013701	.0022027
	Cos 6	.0081971	-.1586503	.0036930	.0023990
	Sin 1	.0520503	.1852949	-.0405273	-.0084972
	Sin 2	-.0269135	-.3424098	.0059050	.0052691
	Sin 3	-.0301697	-.3827283	-.0141138	.0042813
	Sin 4	-.0024391	-.1890056	.0207311	-.0009604
	Sin 5	-.0140458	.1369345	-.0033695	.0007219
	Sin 6	.0108562	.1520455	.0071369	-.0051638
K, 2	Constant	8.2909628	73.2500000	.1784281	-.1040591
	Cos 1	-.1762658	-5.1687097	-.0149049	.0375563
	Cos 2	.0707887	-.8214932	.0429286	-.0141804
	Cos 3	.0578919	.3027581	-.0023471	-.0125446
	Cos 4	-.0074973	.3392090	.0070985	.0029161
	Cos 5	-.0073355	-.0354935	.0013795	.0017729
	Cos 6	.0072606	-.0703854	.0065824	.0019020
	Sin 1	.0471840	.1283946	-.0389161	-.0017633
	Sin 2	-.0301195	-.5498457	-.0104279	.0061491
	Sin 3	-.0115535	-.4734041	-.0140684	.0017381
	Sin 4	-.0015731	-.2253053	.0123246	.0014039
	Sin 5	-.0108186	-.0168161	.0048552	.0008384
	Sin 6	-.0060822	.1248254	.0080072	-.0015754
K, 3	Constant	8.3578268	71.3750000	.1944620	-.0941288
	Cos 1	-.1294272	-5.5714716	.005109	.0347372
	Cos 2	.0751993	-.1557649	.0353104	-.0142468
	Cos 3	.0198924	.3319712	-.0080685	-.0047589
	Cos 4	-.0004569	.0294626	.0058568	.0036599
	Cos 5	.0008590	-.2520228	.0018154	.0020201
	Cos 6	.0076890	-.1792657	.0058214	-.0020462
	Sin 1	.0257314	-.0986995	-.0328608	.0033775
	Sin 2	-.0367547	-.9554075	.0100110	.0099362
	Sin 3	.0048249	-.4965528	.0137095	.0004723
	Sin 4	.0122568	.1502035	.0118110	-.0036687
	Sin 5	-.0027130	.1356316	.0052973	-.0019470
	Sin 6	-.0169765	.0000000	-.0043098	.0020340
K, 4	Constant	8.3971276	70.0708333	.1796257	-.0926478
	Cos 1	-.0748857	-4.5498125	-.0451789	.0155140
	Cos 2	.0628215	.3797725	-.0056263	-.0188644
	Cos 3	.0205632	-.1284269	.0144340	-.0030768
	Cos 4	-.0036976	-.3436211	.0181576	.0017261
	Cos 5	.0088733	.0025403	-.0297343	-.0009343
	Cos 6	-.0012587	.0230968	-.0006281	-.0003616
	Sin 1	.0378818	.2644732	-.0633593	-.0002402
	Sin 2	-.0547457	-.7677660	.0266415	.0111606
	Sin 3	-.0270750	-.1598880	.0061358	.0003837
	Sin 4	.0184307	.0460054	-.0140902	-.0026026
	Sin 5	-.0043656	.0616603	-.0079481	-.0018301
	Sin 6	.0176435	-.5931432	.0461685	.0050041
K, 5	Constant	8.5047969	68.9166667	.1407376	-.0897284
	Cos 1	.0083346	-4.6529356	-.0016084	.0177229
	Cos 2	.0003975	.2982035	-.0138547	-.0005569
	Cos 3	-.0084703	.3985200	-.0089796	-.0016517
	Cos 4	.0282854	-.9717052	.0186920	.0082788
	Cos 5	.0586690	.4004649	-.0135471	-.0087578
	Cos 6	-.0029402	-.4207092	.0008712	.0016466
	Sin 1	.0997451	-1.1825856	.0201719	.0073214
	Sin 2	.0516563	-1.4195980	.0239045	.0095884
	Sin 3	.0128657	-.2180476	.0259267	.0004367
	Sin 4	.0177685	.0531464	-.0127650	-.0009010
	Sin 5	-.0263897	.5072412	-.0010284	-.0013352
	Sin 6	.0337555	-.5482702	-.0008266	.0035559
K, 6	Constant	8.6517426	66.8229167	.1409136	-.0749910
	Cos 1	-.0370451	-4.9876325	.0046305	.0159613
	Cos 2	-.0343524	1.0101195	-.0171867	.0008709
	Cos 3	-.0218729	-1.2344040	-.0126698	.0038346
	Cos 4	.0598460	-.9568701	.0302579	.0030216
	Cos 5	.0355728	-.0705667	-.0060640	-.0023803
	Cos 6	.0345584	-.3404169	-.0015417	.0005726
	Sin 1	.1084300	.5693895	-.0023530	-.0064361
	Sin 2	-.0146683	.0367430	.0054035	.0017862
	Sin 3	-.0369367	-.4429932	.0174821	.0000208
	Sin 4	.0076919	-.1803368	-.0096574	.0033954
	Sin 5	-.0577212	.8665934	-.0218716	-.0001447
	Sin 6	.0203793	-.7063537	-.0000407	.0016903

TABLE 6. Epstein-Fourier Coefficients for Hall Conductivity

		max value	max latitude	up-slope	down-slope
K, 0	Constant	2.7583333	70.9270833	.5430208	-.6710083
	Cos 1	-.5603526	-1.4479718	-.3334788	.2535236
	Cos 2	.7095907	-.7911263	.2223202	-.1569332
	Cos 3	.0604564	-.1704330	.0203252	-.0519584
	Cos 4	.1942398	1.0007805	.0388095	-.1086571
	Cos 5	.1655698	-.0404724	.1027042	.0169650
	Cos 6	.0791731	-.9711485	-.0976908	.0640061
	Sin 1	.5283606	-1.5744713	.1672962	.2117994
	Sin 2	-.1088354	-1.1907920	-.0276489	-.1326264
	Sin 3	-.1276641	.1657324	-.0925705	.1271353
	Sin 4	-.2912483	.4462761	-.0285336	.0286423
	Sin 5	-.3695270	-.6442632	.0830744	.1859250
	Sin 6	-.0735649	-.0981719	-.0663541	-.0236317
K, 1	Constant	4.8360417	69.6250000	1.0627021	-.9502937
	Cos 1	.8434215	-1.8818600	.2952033	-.2177987
	Cos 2	.4354745	-.2942536	.0748335	.0016742
	Cos 3	-.2749059	-.4034963	-.1029672	.3031466
	Cos 4	-.1430014	-.0215684	-.1008671	.2112045
	Cos 5	.0553424	-.1111628	-.0477245	-.0793805
	Cos 6	.1852824	-.1042102	.0795808	-.0918663
	Sin 1	1.9244905	-1.3952794	.6840835	-.1842365
	Sin 2	-.4590670	-1.3282438	.1109565	.1427165
	Sin 3	-.0486048	-.0968973	.0335025	.0584180
	Sin 4	.0539422	.0991725	.0764872	-.1026860
	Sin 5	.0655059	-.1104204	.0118614	-.0378415
	Sin 6	.0917029	.0272201	.1042123	-.0297088
K, 2	Constant	7.2189583	69.2187500	1.5508062	-1.3937083
	Cos 1	2.2869975	-2.3106173	.8729207	-.3873089
	Cos 2	.7658015	-.6651852	.3313611	-.0324994
	Cos 3	.0858780	-.3403501	.2991748	.2908475
	Cos 4	-.0504954	-.3165834	.6377540	.1813414
	Cos 5	-.0715295	-.5789755	.3082071	.0516969
	Cos 6	.0325820	.0736875	-.0289491	.0273734
	Sin 1	2.8989121	-1.6590557	.9542872	-.7229911
	Sin 2	-.5101666	-1.0118879	.1121867	-.1986246
	Sin 3	.5489824	-.2062100	.0928020	-.1931953
	Sin 4	.6240107	-.1703270	.2855607	-.4165769
	Sin 5	.6830349	-.0976242	.5309347	-.1005444
	Sin 6	.1596690	.3027230	.2686735	-.1478393
K, 3	Constant	10.0956250	68.5687500	2.3899479	-1.8148187
	Cos 1	4.6945905	-2.5171675	2.1987400	-.5808578
	Cos 2	-.1368916	-1.1420728	.8623940	.3096004
	Cos 3	-.1602042	-.2289998	.6900643	.3766520
	Cos 4	.5540120	-.1730619	.5801013	-.0892347
	Cos 5	.9169015	-.6109952	.3801142	-.1280161
	Cos 6	.1055958	.5079539	.1477341	-.1104187
	Sin 1	3.6216916	-2.0819841	1.3043800	-.9819689
	Sin 2	-.6399891	-1.0413881	.6026834	-.1678922
	Sin 3	1.0016951	.5261618	.5150961	-.3521850
	Sin 4	.5015487	-.2260380	.2940746	-.0533188
	Sin 5	-.0548462	.5490804	.0626488	-.0024139
	Sin 6	.2895085	.1490361	.1097321	-.0880467
K, 4	Constant	10.5539583	67.0520833	2.4154562	-1.7039771
	Cos 1	5.0368212	-1.4380474	1.8651015	-.7921974
	Cos 2	-1.4508298	-.0999283	-.1286010	.1827648
	Cos 3	-.3386978	-.2256838	.2897210	.1455383
	Cos 4	.7145481	.7854492	.4342100	-.1825706
	Cos 5	1.0521712	-.6754360	.1643043	-.1776439
	Cos 6	.3106666	.5504117	-.0377262	.0351027
	Sin 1	3.7795599	-.4695603	.8035522	-1.0630104
	Sin 2	-2.2036197	-1.4263343	-.0061968	.0007466
	Sin 3	-.4038805	1.0786975	-.0381278	-.1379479
	Sin 4	.1877177	-.3410563	-.1786102	-.0955437
	Sin 5	.3821918	.1441355	-.0286417	-.2698556
	Sin 6	.7227088	.1732274	.3489599	-.2086555
K, 5	Constant	9.7808333	67.5479167	1.2953417	-1.2377000
	Cos 1	4.0828747	-.8252280	.3205802	-.4608375
	Cos 2	-3.0447562	.9037982	-.7337893	.3273742
	Cos 3	-1.3303617	.5731158	-.2707783	.1119489
	Cos 4	-.2996340	.2322553	.0924913	-.0338068
	Cos 5	.5788535	-.2093377	.0271088	-.0558071
	Cos 6	.2709488	.5551483	-.0459448	.0901108
	Sin 1	2.5820114	-1.1411468	.3528058	-.6403901
	Sin 2	-2.1114761	.0118324	-.4490833	.1414125
	Sin 3	-.4143120	.2220543	-.1697168	.0110541
	Sin 4	.2391930	-.3030228	.2391388	-.0583405
	Sin 5	.5039429	.0011819	.1940858	-.1483512
	Sin 6	-.1333369	-.2611443	-.0800368	-.0755900
K, 6	Constant	13.9289583	65.0187500	2.0588917	-2.1786729
	Cos 1	10.1588173	-2.4207529	1.4649824	-.9497801
	Cos 2	-2.0789625	-1.5525052	-.6865596	.5574923
	Cos 3	-.4669589	.3268133	-.0568233	-.3584154
	Cos 4	.9080207	-.2252280	.1397604	-.0551213
	Cos 5	.4411382	-.5814691	-.2259140	.2615956
	Cos 6	.0918056	.6422800	-.1596379	.0511246
	Sin 1	4.0465757	-.8253996	.9358234	-.6483012
	Sin 2	.1479905	-.2498451	.1395212	-.8750918
	Sin 3	.4186253	1.4356406	-.1696164	-.5047729
	Sin 4	-.1309743	-1.5516421	-.0525527	.2756927
	Sin 5	.1295592	.9542121	.0239853	-.4123370
	Sin 6	.3858664	-.4469092	.1524355	-.2234162

TABLE 7. Epstein-Fourier Coefficients for Pedersen Conductivity

		max value	max latitude	up-slope	down-slope
K, 0	Constant	1.9752083	71.7708333	.5717104	-.2543417
	Cos 1	.1130936	-2.2547427	.2175173	-.0453745
	Cos 2	.3619121	-1.1885288	.1759850	-.0710386
	Cos 3	.1148692	.8967821	-.0320229	-.0265617
	Cos 4	.1334866	.0952250	-.0744442	-.0462907
	Cos 5	.0728649	-.4799590	.0146303	-.0023734
	Cos 6	.0020049	.0552412	.0828215	.0190039
	Sin 1	.4995522	-1.6743230	.1537368	-.1215096
	Sin 2	-.1902356	-.1058221	.0716203	.0387746
	Sin 3	-.0952397	.0437346	-.0137312	.0308323
	Sin 4	-.1536157	-.4250950	-.0814600	-.0029734
	Sin 5	-.1402175	.5415827	-.1238568	-.0177773
	Sin 6	-.0439370	.1042100	-.1717793	-.0314532
K, 1	Constant	3.1712500	70.7500000	.7165729	-.4452604
	Cos 1	1.0460200	-2.9196623	.4346830	-.1815297
	Cos 2	.1379087	-.7150610	.0467664	.0052956
	Cos 3	-.0299421	.3733959	-.0917933	.0789029
	Cos 4	-.0749650	-.2896231	.0098491	.0802530
	Cos 5	-.0013665	-.2592726	.0242828	.0635638
	Cos 6	.1106588	.1088801	.0139084	-.0430246
	Sin 1	.8085247	-1.8355858	.2766084	-.0593998
	Sin 2	-.4091995	-.5711044	.0446986	.1262351
	Sin 3	.1050777	-.1693432	.1006073	.0165674
	Sin 4	-.0233221	-.3406539	.0850947	.0306373
	Sin 5	.0288914	.1691063	.0326972	-.0242391
	Sin 6	.0627288	-.0000000	.0436159	-.0517958
K, 2	Constant	4.3137500	70.1458333	.8209917	-.6039646
	Cos 1	1.9275469	-3.2625586	.4892092	-.2107561
	Cos 2	.0964910	-.2705674	-.0435020	.0159070
	Cos 3	.1477087	.0958508	.1088144	.1274871
	Cos 4	-.1099220	-.6248849	.2465228	.0935024
	Cos 5	-.0332367	-.1556599	.1520316	.0343090
	Cos 6	.0956958	-.0000002	.0953473	-.0575092
	Sin 1	1.0182152	-1.7175094	.2547495	-.2152703
	Sin 2	-.2810904	-.7246931	.0788403	.0275124
	Sin 3	.3487381	-.4877543	.0806739	-.1309935
	Sin 4	.1427723	-.2508208	.0395550	.0098114
	Sin 5	.2053887	.0474664	.1410467	.0260413
	Sin 6	.0478526	-.0450997	.1486745	.0083321
K, 3	Constant	5.3681250	69.6000000	1.0638625	-.7238229
	Cos 1	2.8279962	-3.7821781	.8834478	-.1338280
	Cos 2	-.3187805	-.4009589	.2099942	.1314392
	Cos 3	.1640319	-.1429645	.2841189	.0694481
	Cos 4	.1400917	-.4882003	.1719627	-.0795906
	Cos 5	.2006710	.1241429	.0868618	-.1167266
	Cos 6	-.0046307	-.2483964	.0621159	.1370907
	Sin 1	.9114709	-1.9005839	.2932267	-.0445403
	Sin 2	-.3668006	-.4507286	.2127840	.0836998
	Sin 3	.5416129	-.0321750	.1145039	-.2242315
	Sin 4	.1796916	.2932245	.0013498	-.0529602
	Sin 5	.0442166	.2336574	.0048137	-.0164184
	Sin 6	.0762248	-.0668093	-.0120836	.0250792
K, 4	Constant	5.5833333	68.7708333	.8618917	-.6730812
	Cos 1	2.9881179	-3.8715167	.5861567	-.1338858
	Cos 2	-.6395703	.4692158	-.1440779	.1079669
	Cos 3	.0407904	.4348478	-.0494269	-.0464855
	Cos 4	.2289197	-.4089190	.0321022	-.0204007
	Cos 5	.1514792	.5260999	-.0304234	-.1015419
	Cos 6	.0188512	-.2807399	-.0948005	.0474771
	Sin 1	1.0235213	-1.5562133	.2829520	-.1686421
	Sin 2	-1.0883883	.2823293	-.0371388	.1360288
	Sin 3	.0821190	-.5886884	-.0303181	-.0059638
	Sin 4	.1444900	.4746813	-.1167050	.0097083
	Sin 5	.1979488	.2069094	-.0098709	-.1002859
	Sin 6	.3237463	-.2581899	.0579316	-.0459037
K, 5	Constant	5.7097917	67.9583333	.6400104	-.5891208
	Cos 1	2.7157215	-2.1349113	.2869028	-.1882091
	Cos 2	-1.5537595	.4418344	-.2290677	.1826844
	Cos 3	-.5065092	-.1738927	-.0529559	.0802309
	Cos 4	-.1512489	-.3809010	.0908214	.0307426
	Cos 5	.2672217	-.1577935	.0854744	-.0821933
	Cos 6	.0343473	-.3491907	.0262790	.0756556
	Sin 1	.8486314	-.8394362	.1559990	-.1708149
	Sin 2	-.4712352	-.1761762	-.1165499	.0839000
	Sin 3	.0009869	.1165620	-.0363228	-.0468929
	Sin 4	.1603752	-.4240373	.1485710	.0053801
	Sin 5	.0917441	-.3613390	.0929087	.0116085
	Sin 6	-.1483726	-.7022501	.0059449	.0570071
K, 6	Constant	7.4172917	66.4729167	.9703229	-.7516958
	Cos 1	5.1378548	-4.5011062	.6465420	-.3948450
	Cos 2	-1.1244043	.0542128	-.3786114	.1043948
	Cos 3	-.0568033	-.3201861	-.1429368	.0603873
	Cos 4	.7238344	-.7941542	.2095081	-.0394841
	Cos 5	.4950576	-.1298506	.1466138	-.0799316
	Cos 6	.1960477	.0437591	-.0688452	.0538642
	Sin 1	1.3627671	-1.1403526	.2885864	-.3226968
	Sin 2	-.4804401	1.3588885	.0097264	-.1102806
	Sin 3	.2477993	-.3788762	-.0096195	-.1443434
	Sin 4	-.0053384	-.2815735	-.0565773	.0192470
	Sin 5	-.0993957	.1621495	-.0179974	-.0414677
	Sin 6	.2929164	-.0907934	.0712223	-.1096790

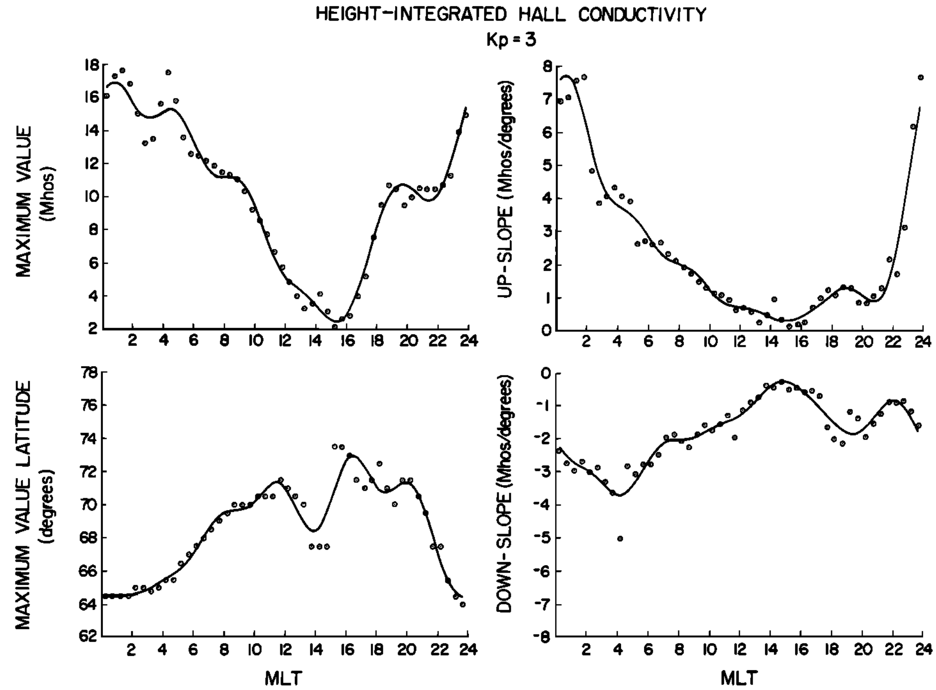

Fig. 4. The Fourier expansion fit to the magnetic local time variation in the four constants of the Epstein function for the height-integrated Hall conductivity for $Kp = 3$.

The quality of the fitting is shown in Figures 6 and 7. In Figures 6 and 7 the original data and the fits are shown for the Hall and Pedersen conductivities, for $Kp = 3$, and for four MLTs. The fitting is of roughly equal quality at all local times. For both the Hall and Pedersen conductivities the fit value is set equal to zero below the oval even though the original data produce values of a fraction of a mho at these latitudes. A careful examination of the original data has shown that these values result from a small level of residual background in the data due to penetrating particles from the radiation belts. These values are therefore not real. Figures 6 and 7 show that the model fits the data most poorly at high latitudes where the data decrease more gradually to the value in the polar cap region than the model.

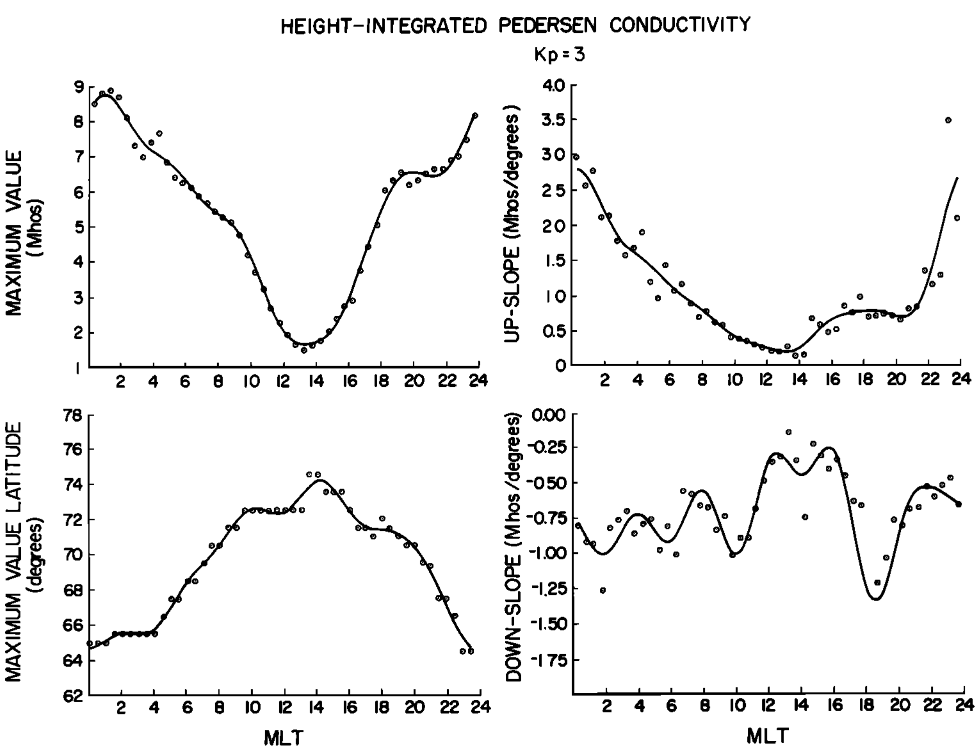

Fig. 5. Same as Figure 4 for the height-integrated Pedersen conductivity.

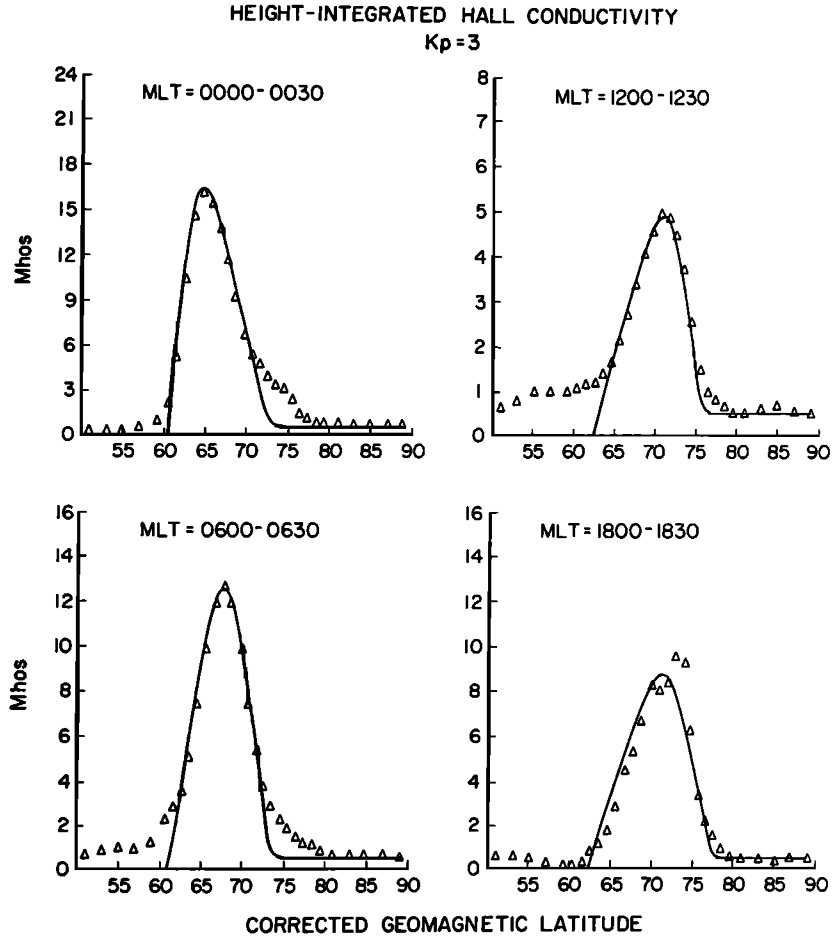

Fig. 6. The values of the calculated height-integrated Hall conductivity (triangles) and the fit values (solid line) for $Kp = 3$ along the midnight meridian (upper left), the dawn meridian (lower left), the dusk meridian (lower right), and the noon meridian (upper right).

A more quantitative measure of how well the fits represent the data is given in Figures 8 and 9. In Figures 8 and 9 the normalized distribution of differences between the original data and the fit are shown for each of the four quantities and for the $Kp = 2$ and $Kp = 5$ cases. Figures 8 and 9 show that the conductivities are best fit by the procedure. For the Pedersen (Hall) conductivity the full width at half maximum of the distribution is approximately 0.16 (0.12) for the $Kp = 2$ case and 0.20 (0.32) for the $Kp = 5$ case. For the integral energy flux and integral number flux, by contrast, the values are 0.32 (0.16) and 0.40 (0.40), respectively. The larger spread in differences for the integral number flux and the integral energy flux is due principally to the fits having been performed to the logarithmic values while the difference distribution is for the linear values. The increase in the widths of the distribution with increasing activity is a general characteristic of the procedure and is the result of increased variability in the data used to construct the averages as Kp increases and a decrease in the number of samples used to determine the average. In general, the distributions are roughly symmetric about 0.

For all four quantities there is a more prominent wing to the distribution toward positive values than toward negative values. These wings have two sources. First, they are the result of the small number of noise spikes remaining in the original data. Second, as shown in Figures 6 and 7, the conductivities have a gradual roll-off for latitudes well above and below the peak. In this roll-off region the Epstein functions provide a poorer fit. The noise spikes have values significantly exceeding the smooth curve produced by the surrounding values. The surrounding values dominate in determining the fit. The contribution of the spikes to the wings does not therefore represent a failing of the fitting since in these cases the fit actually provides a better value than the original data.

5. Discussion and Conclusion

In this section we have two aims: (1) to compare the values of the conductivities produced by electron precipitation to those produced by solar radiation and (2) to compare the conductivity results from this paper to previous work using particle data.

The importance of the height-integrated conductivities produced by electron precipitation to the dynamics of the auroral oval is dependent, in part, on their magnitude relative to the conductivity produced by solar radiation. A number of determinations of the solar contribution to ionospheric conductivities have been made using incoherent radar data [*Mehta*, 1979; *Senior*, 1980; *Vickrey et al.*, 1981; *de la Beaujardiere et al.*, 1982; *Robinson and Vondrak*, 1984]. *Robinson and Vondrak* [1984] showed that the values of the solar radiation induced, height-integrated, Hall and Pedersen conductivities

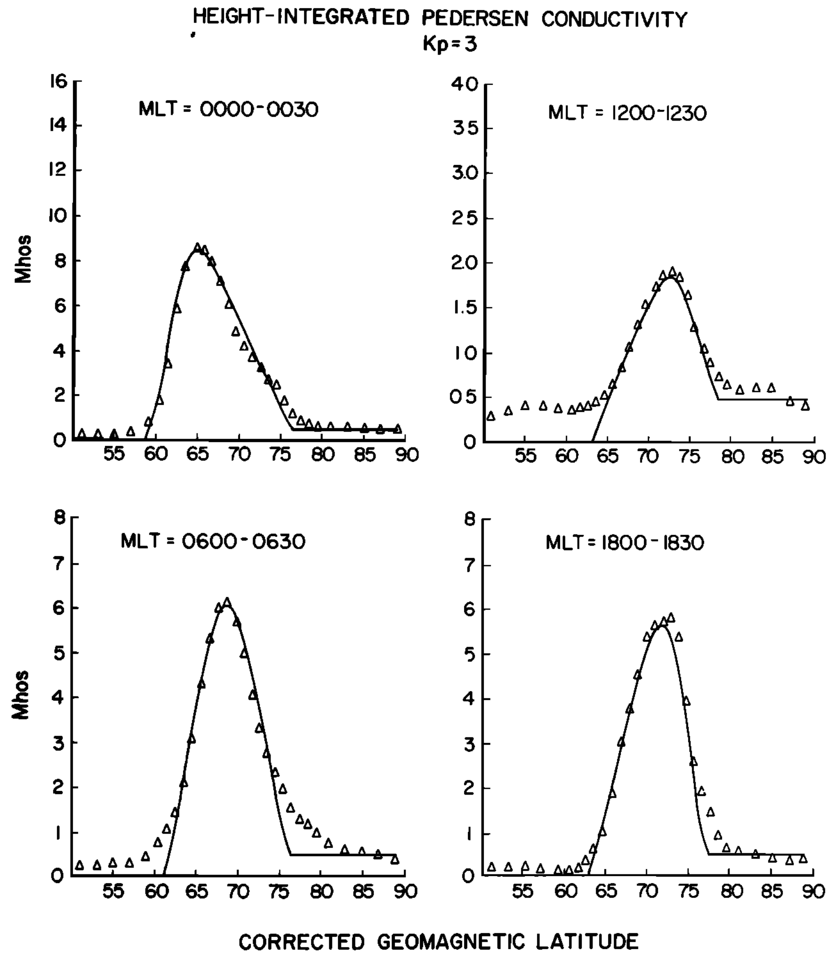

Fig. 7. Same as Figure 6 for the Pedersen conductivities for $Kp = 3$.

are dependent on both the level of the solar power flux at 10.7 cm, S_a, and the solar elevation angle θ. The dependence is given by

$$\Sigma_P = 0.88\ (S_a \cos\theta)^{0.5} \tag{6}$$

$$\Sigma_H = 1.5\ (S_a \cos\theta)^{0.5} \tag{7}$$

Since over a solar cycle, the monthly average of the solar flux S_a varies by roughly a factor of 4 from approximately 60 at solar minimum to 240 at solar maximum, the height-integrated conductivities will vary by a factor of 2. At solar minimum the equations for the conductivities would be

$$\Sigma_P = 6.8 \cos^{0.5}\theta \quad \text{mhos} \tag{8}$$

$$\Sigma_H = 11.6 \cos^{0.5}\theta \quad \text{mhos} \tag{9}$$

In Table 8 the relative contributions to the conductivities from solar radiation and electron precipitation are shown. The comparison is made at the position of the maximum, electron-produced conductivity for $Kp = 2$ at noon local time. For the solar-produced conductivity the diurnal extremes at this position (72° CGL) were calculated at the summer solstice and at equinox. Values in parentheses give the ratio of the electron to solar-produced conductivity. At the summer solstice, where the solar conductivity should be greatest, the ratio varies diurnally between 0.40 (0.57) and 0.49 (0.59) for the Pedersen (Hall) conductivity. At the equinox the ratios vary between 0.53 (0.74) and 1.04 (1.50). For periods closer to the winter solstice the ratios become larger. At solar maximum the ratios would decrease by a factor of 2 assuming that the level of precipitation at a fixed level of activity does not have a solar cycle dependence. The above illustrates that during all seasons of the year, the Hall and Pedersen conductivities due to electron precipitation are significant relative to that from solar radiation, at least at or near the peak of the precipitating electron produced conductivities on the dayside of the oval.

Our results on the Hall and Pedersen conductivities can be compared to those *Wallis and Budzinski* [1981] and *Spiro et al.* [1982]. In those papers the conductivities were derived in a similar manner using precipitating electron data. Wallis and

TABLE 8. Electron- and Solar-Produced Conductivity

		Solar Minimum, mhos			
		Summer Solstice		Equinox	
	Electrons, mhos	Diurnal Minimum	Diurnal Maximum	Diurnal Minimum	Diurnal Maximum
Σ_P	2.5	5.1 (0.49)	6.2 (0.40)	2.4 (1.04)	4.7 (0.53)
Σ_H	6.0	8.7 (0.69)	10.6 (0.57)	4.0 (1.50)	8.1 (0.74)

Values in parentheses give ratio of electron- to solar produced conductivity.

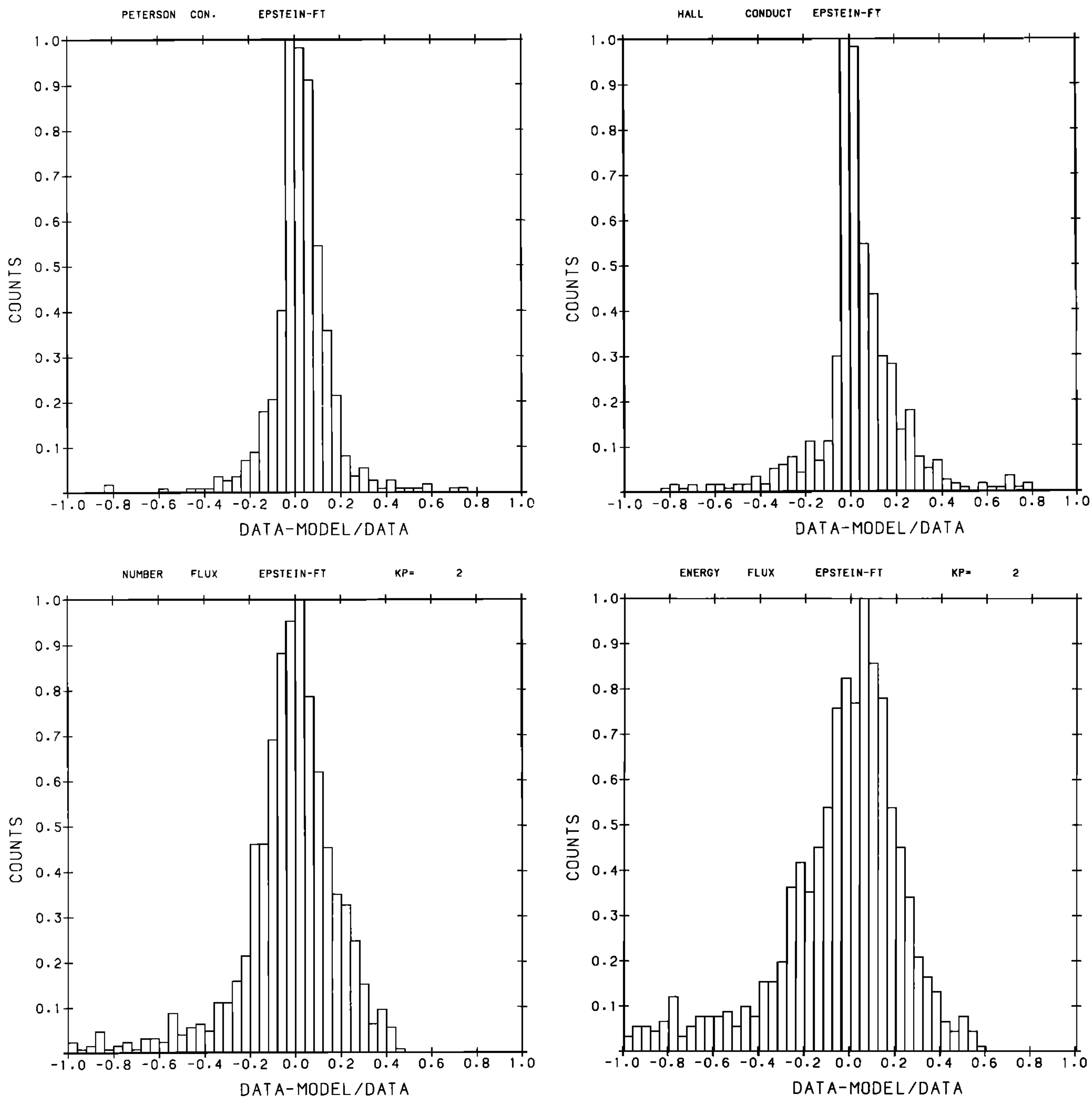

Fig. 8. The distribution of percentage differences between the calculated and fit values of the height-integrated Pedersen conductivity, the height-integrated Hall conductivity, the integral number flux, and the integral energy flux for $Kp = 2$. The percentage difference is calculated by subtracting the fit value for each of the four quantities from the original value of the data and then dividing by the original value. The counts refer to the number of CGL-MLT bins in which a given percentage difference was observed. The counts have been normalized to the largest value of this number.

Budzinski's results were derived using particle data from the ISIS satellite. The average number flux of electrons at four energies (0.15, 1.27, 9.65, and >22 keV) was determined for a spatial grid in magnetic local time and invariant latitude and for two ranges of activity as measured by Kp ($0 < Kp \leq 3$, and $3 < Kp < 9$). These fluxes were the input to a code for calculating the height ionization profile. These profiles were then combined with an atmospheric model to calculate the height variable conductivities and the height-integrated conductivities. Their final conductivities also included the effects of galactic EUV and other backgrounds. In the work by *Spiro et al.* [1982], data from the AE-C and AE-D satellites were used to determine the average value of the integral energy flux and average energy for precipitating electrons over the energy range from 200 eV to 27 keV in each element in a spatial grid in magnetic local time–geomagnetic latitude coordinates. The pattern was determined for four levels of activity as determined by AE. These values were then used to calculate in each spatial element the height-integrated Hall and Pedersen conductivities as in our work.

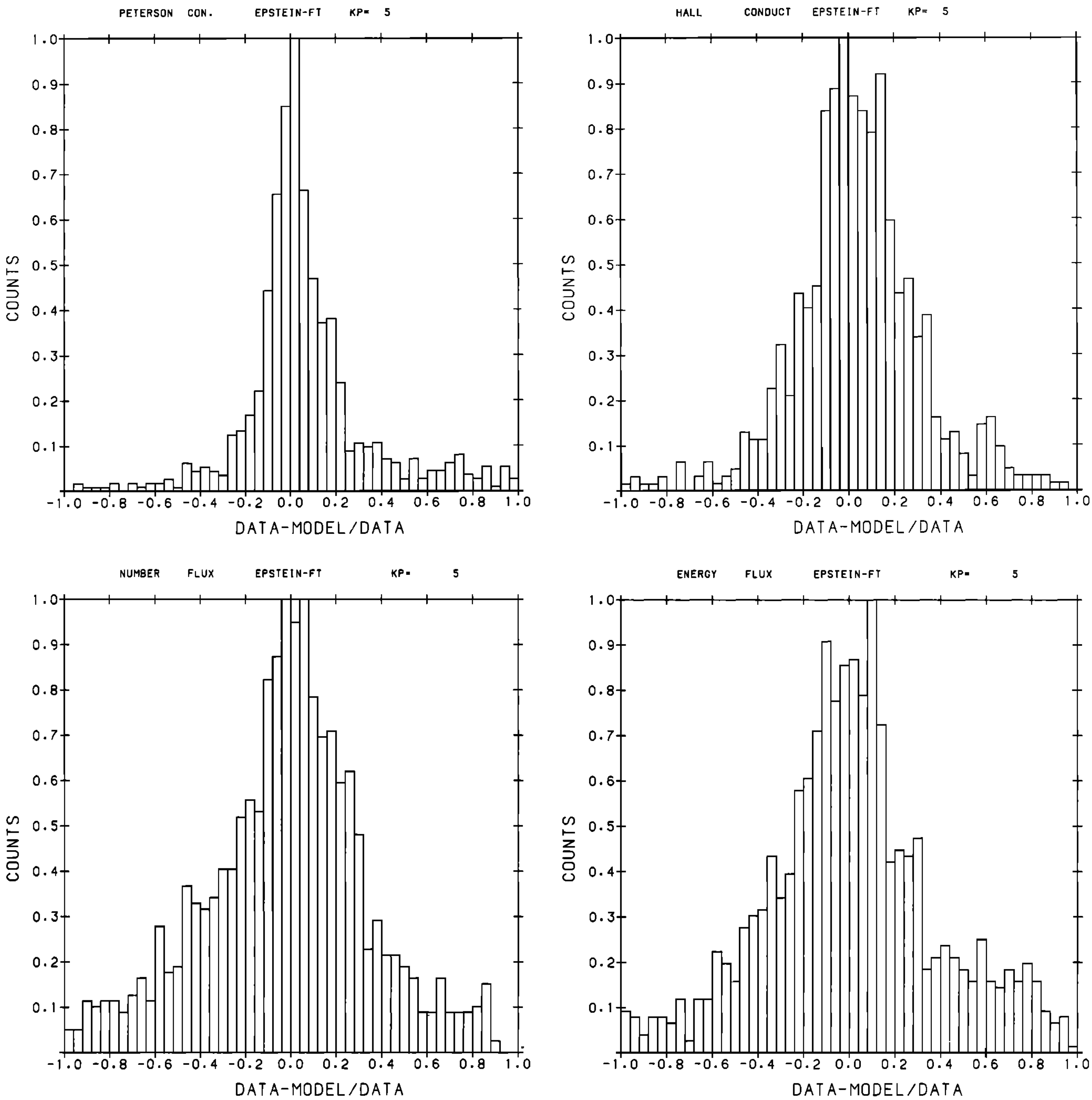

Fig. 9. Same as Figure 10 for $Kp = 5$.

A comparison of the Wallis and Budzinski results with those from our study are shown in Figure 10. In Figure 10 the Hall and Pedersen conductivities for the Wallis and Budzinski low-activity case are plotted along with the conductivities from our study for $Kp = 2$ for the noon-midnight and dawn-dusk meridians. The Wallis and Budzinski conductivities have consistently higher peak values than those in our study. This is the result of Wallis and Budzinski using the incorrect collision frequency coefficient [*Vickrey et al.*, 1981; *Spiro et al.*, 1982]. *Vickrey et al.* [1981] have pointed out that this can lead to an overestimate of the height-integrated conductivities of from 5 to 30%. Taking this into account, the conductivities from the two studies are in good agreement. The best agreement is on the morningside and at noon where the shape and the location of the peaks for the two curves are very similar. The agreement in the midnight to dusk sector is poorer. In this region the Wallis and Budzinski pattern is shifted poleward of the pattern of the present study. This shift is most likely the result of underestimating the conductivity due to the limited spectral information available to Wallis and Budzinski. Such an underestimation would be greatest on the eveningside where the electron spectrum is colder.

Even with the correction to the work of *Spiro et al.* [1982], our agreement with their results is poor both in the spatial extent and the level of the conductivities. The poor agreement results principally from differences in the latitudinal and local

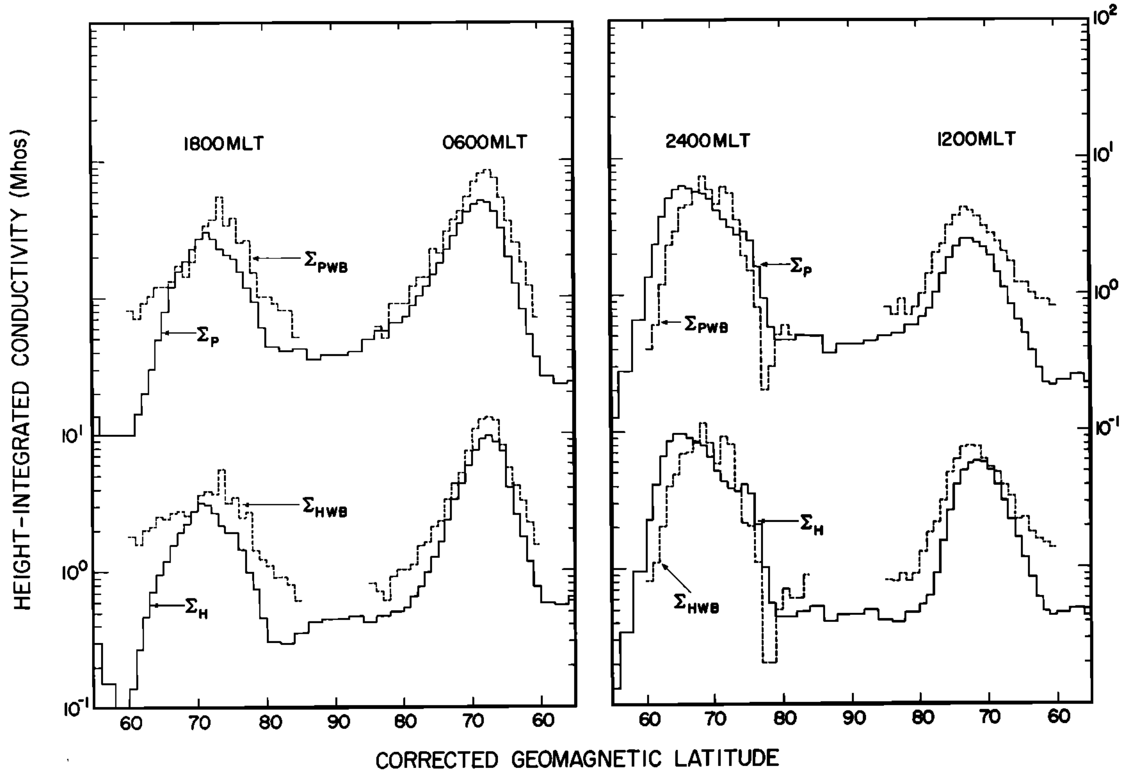

Fig. 10. The latitudinal profiles of the height-integrated Hall and Pedersen conductivities along the noon-midnight and dawn-dusk meridians for the low activity case of *Wallis and Budzinski* [1981] and for the $Kp = 2$ case from this study. The Wallis and Budzinski results are plotted as a dashed line, and the results from the present study are plotted as a solid line.

time pattern of the integral energy flux used in the two calculations. In Figure 11 we have plotted the integral energy flux, as reported by the two studies, along the noon-midnight and dawn-dusk meridians. For *Spiro et al.* [1982] the two curves are for the $100 < AE < 300$ gamma and the $300 < AE < 600$ gamma cases. These are compared to the $Kp = 2$ and $Kp = 4$ cases, respectively, from the present work.

For the lower-activity case the latitudes of the peak values at midnight, 0600, and noon for the Spiro et al. results are poleward of those in our study. At 1800 the latitude of the peak value from Spiro et al. is shifted equatorward of the latitude for the peak in this study. At all four local times the peak values from Sprio et al. are larger than in the present study. If we compare the Spiro et al.. results to our results for $Kp < 2$, the location of the peaks is in better agreement, but the difference in the peak values is even greater. For the higher-activity case the peak values from dawn to dusk through midnight in the two studies are in good agreement, but the Spiro et al. profiles are much broader in latitudinal extent. At midnight, in the Spiro et al. study, the integral energy flux remains above 1 erg/cm^2 s up to 78°, while we show the integral energy flux falling below this level at 71°. At noon the peak from the Spiro et al. results is shifted poleward of that for our study, and the energy flux at the peak is much larger. Also, in our study, at all local times except noon, the peak energy flux increases with increasing Kp, while for Spiro et al. the integral energy flux at midnight decreases with increasing AE for the two cases shown. Comparing our results at higher or lower Kp to the Spiro et al. results does not produce any better agreement.

The reason for the large discrepancy between the two models is unclear. Part of it is likely the result of the smaller size of the data set available to Spiro et al. This is supported by the fact that the Spiro et al. results vary less smoothly with latitude than the results of our study. Part of the difference might be the result of using AE in one case and Kp in the other. Kp is a 3-hour index, and therefore our results represent an average over all substorm phases occurring at a fixed value of Kp. By using an AE sort, Spiro et al. may be separating different phases of the substorm which may directly order the width of the oval. This could account for their producing a broader oval at higher AE than we see at high Kp.

In conclusion, we have determined the global pattern of the height-integrated Hall and Pedersen conductivities produced by precipitating auroral electrons for seven levels of geomagnetic activity as measured by Kp. These conductivites are found to vary smoothly in magnetic local time and corrected geomagnetic latitude; typically displaying a single peak conductivity value at each MLT. There are generally two maxima for the conductivities in the nightside auroral oval: one near midnight and another several hours postmidnight. The larger of the two maxima is at midnight. The peak Hall and Pedersen conductivity decreases with MLT away from midnight on both the morningside and eveningside of the oval with the lowest peak conductivities seen in the postnoon sector. At noon, and over much of the dayside, the peak conductivity increases with increasing Kp up to $Kp = 2$ and is either flat or slightly decreasing for higher Kp. The latitudinal profile of the conductivities varies principally in level and not in shape with activity and at most local times is not symmetric in latitude about the peak value. For activity above $Kp = 0$ the ratio of Hall to Pedersen conductivity is greatest on the morningside

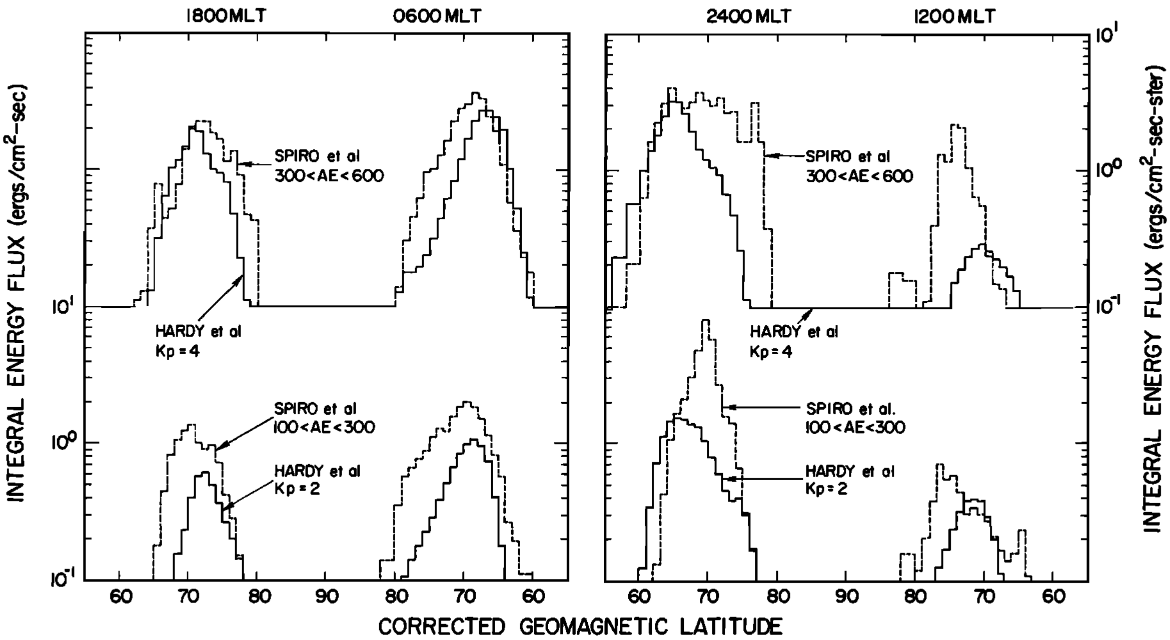

Fig. 11. The latitudinal profiles of the integral energy flux along the noon-midnight and dawn-dusk meridians from *Spiro et al.* [1982] for *AE* in the ranges from 100 to 300 gammas and from 300 to 600 gammas and for $Kp = 2$ and $Kp = 4$ from this study.

of the oval and at noon. On the eveningside the Hall and Pedersen conductivities are approximately equal at all latitudes. At all local times the ratio of Hall to Pedersen conductivity is close to one in the poleward portion of the oval. For $Kp = 0$ the highest conductivities are found on the eveningside, and the ratio of the Hall to Pedersen conductivity at the peak on the morningside is less than 1. The results are found to be in approximate agreement with the work of *Wallis and Budzinski* [1981] but to differ significantly from the results of *Spiro et al.* [1982]. The peak conductivity from precipitating electrons on the dayside is found to be significant relative to the conductivity due to solar radiation.

The global patterns for the height-integrated Hall and Pedersen conductivity as well as the integral energy flux and the integral number flux were fit using both a spherical harmonic expansion and Epstein functions. The Epstein functions were found to reproduce the data better. Each map at each level of activity can be represented using a total of 52 coefficients for the Fourier expansion in MLT of the four quantities in the Epstein function. For $Kp = 2$ the normalized distribution of differences between the original data and the model for the Hall (Pedersen) conductivity has a full width at half maximum of 0.12 (0.16) and for the integral energy (number) flux a value of 0.32 (0.16). The widths of these distributions tend to increase for increasing and decreasing activity.

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